

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	3.2	0.5	0	0	85.6
U2 - Beta(0.5, 0.5)	11.5	2.8	0.2	0	0	85.5
U2 - Beta(0.7, 0.3)	11.3	2.7	0.2	0.1	0	85.7
U2 - Beta(1, 1)	11.2	2.8	0.3	0.1	0	85.6
U3 - Beta(0.3, 0.7)	11.4	2.7	0.2	0	0	85.7
U3 - Beta(0.5, 0.5)	11.7	2.3	0.2	0	0	85.8
U3 - Beta(0.7, 0.3)	11.6	2.7	0.4	0	0	85.3
U3 - Beta(1, 1)	11.7	2.4	0.2	0	0	85.7
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.4	1	0.2	0	-
U2 - Beta(0.5, 0.5)	11.7	2.4	1	0.2	0	-
U2 - Beta(0.7, 0.3)	11.7	2.4	0.9	0.2	0	-
U2 - Beta(1, 1)	11.8	2.4	0.9	0.2	0	-
U3 - Beta(0.3, 0.7)	11.8	2.3	1	0.2	0	-
U3 - Beta(0.5, 0.5)	11.7	2.4	1	0.2	0	-
U3 - Beta(0.7, 0.3)	11.8	2.4	0.9	0.2	0	-
U3 - Beta(1, 1)	11.7	2.4	0.9	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	1.4	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	3.5	1.3	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	3.5	1.4	0.3	0.1	0	-
U2 - Beta(1, 1)	3.5	1.4	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	3.5	1.3	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	3.5	1.3	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	3.5	1.4	0.3	0.1	0	-
U3 - Beta(1, 1)	3.5	1.4	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						543.3
U2 - Beta(0.5, 0.5)						543.2
U2 - Beta(0.7, 0.3)						541.9
U2 - Beta(1, 1)						543
U3 - Beta(0.3, 0.7)						543.2
U3 - Beta(0.5, 0.5)						542.6
U3 - Beta(0.7, 0.3)						542.8
U3 - Beta(1, 1)						542.2
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	41.8	22.3	27.3	5.5	0.5	2.6
U2 - Beta(0.5, 0.5)	42.1	22.5	27.3	5	0.5	2.6
U2 - Beta(0.7, 0.3)	42.3	22.9	26.7	5	0.5	2.6
U2 - Beta(1, 1)	42.3	21.7	27.7	5	0.7	2.6
U3 - Beta(0.3, 0.7)	44	21.9	26.5	4.3	0.6	2.7
U3 - Beta(0.5, 0.5)	45	22.5	24.9	4.4	0.5	2.7
U3 - Beta(0.7, 0.3)	45.4	22.8	24.9	4.1	0.3	2.5
U3 - Beta(1, 1)	45.7	21.7	25.1	4.5	0.5	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	13.5	6	6.4	2.7	0.9	-
U2 - Beta(0.5, 0.5)	13.6	6	6.3	2.7	0.8	-
U2 - Beta(0.7, 0.3)	13.6	6	6.4	2.7	0.8	-
U2 - Beta(1, 1)	13.6	5.9	6.4	2.7	0.9	-
U3 - Beta(0.3, 0.7)	13.4	6	6.5	2.6	0.9	-
U3 - Beta(0.5, 0.5)	13.6	6	6.3	2.7	0.8	-
U3 - Beta(0.7, 0.3)	13.6	6.1	6.4	2.7	0.8	-
U3 - Beta(1, 1)	13.6	6	6.4	2.7	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	2.8	1.8	0.7	0.2	-
U2 - Beta(0.5, 0.5)	4.1	2.8	1.8	0.7	0.2	-
U2 - Beta(0.7, 0.3)	4.1	2.8	1.8	0.7	0.2	-
U2 - Beta(1, 1)	4.1	2.7	1.8	0.7	0.2	-
U3 - Beta(0.3, 0.7)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(0.5, 0.5)	4.1	2.8	1.8	0.8	0.2	-
U3 - Beta(0.7, 0.3)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(1, 1)	4.1	2.7	1.8	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1093.1
U2 - Beta(0.5, 0.5)						1092.7
U2 - Beta(0.7, 0.3)						1092.6
U2 - Beta(1, 1)						1093.1
U3 - Beta(0.3, 0.7)						1093.1
U3 - Beta(0.5, 0.5)						1093.2
U3 - Beta(0.7, 0.3)						1092.6
U3 - Beta(1, 1)						1093.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.3	41.4	16.4	4.2	0.5	18.2
U2 - Beta(0.5, 0.5)	19	41.3	17.3	3.6	0.5	18.3
U2 - Beta(0.7, 0.3)	19.1	42.2	16.4	3.5	0.6	18.2
U2 - Beta(1, 1)	19.3	42.2	15.3	4.5	0.5	18.2
U3 - Beta(0.3, 0.7)	19.6	43.9	14.7	3.6	0.2	18
U3 - Beta(0.5, 0.5)	20.5	44.1	13.5	3.5	0.3	18.1
U3 - Beta(0.7, 0.3)	20.5	43.7	14.2	3.2	0.3	18.1
U3 - Beta(1, 1)	20.8	43.1	14	3.7	0.3	18.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	7.9	5.4	2.5	0.8	-
U2 - Beta(0.5, 0.5)	10.4	8	5.3	2.5	0.8	-
U2 - Beta(0.7, 0.3)	10.4	8.1	5.3	2.4	0.8	-
U2 - Beta(1, 1)	10.4	8.1	5.2	2.5	0.8	-
U3 - Beta(0.3, 0.7)	10.4	8	5.4	2.4	0.8	-
U3 - Beta(0.5, 0.5)	10.4	8.1	5.2	2.5	0.8	-
U3 - Beta(0.7, 0.3)	10.5	8	5.3	2.5	0.8	-
U3 - Beta(1, 1)	10.4	8	5.4	2.4	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	3.4	1.5	0.8	0.2	-
U2 - Beta(0.5, 0.5)	3.1	3.4	1.5	0.8	0.2	-
U2 - Beta(0.7, 0.3)	3.1	3.5	1.5	0.8	0.2	-
U2 - Beta(1, 1)	3.1	3.5	1.5	0.8	0.3	-
U3 - Beta(0.3, 0.7)	3	3.4	1.5	0.8	0.2	-
U3 - Beta(0.5, 0.5)	3.1	3.4	1.5	0.8	0.2	-
U3 - Beta(0.7, 0.3)	3.1	3.4	1.5	0.8	0.2	-
U3 - Beta(1, 1)	3.1	3.5	1.5	0.8	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						988.3
U2 - Beta(0.5, 0.5)						987.2
U2 - Beta(0.7, 0.3)						986.8
U2 - Beta(1, 1)						987.2
U3 - Beta(0.3, 0.7)						988.1
U3 - Beta(0.5, 0.5)						987.3
U3 - Beta(0.7, 0.3)						986.7
U3 - Beta(1, 1)						987.5
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.5	11.8	29.1	39.4	7.8	4.4
U2 - Beta(0.5, 0.5)	7.4	12.3	27.2	40.5	8	4.6
U2 - Beta(0.7, 0.3)	8.1	12.5	27.6	39.2	8	4.6
U2 - Beta(1, 1)	8.8	11.8	26.6	40.2	8.2	4.4
U3 - Beta(0.3, 0.7)	11.6	11.6	26.7	38.6	7	4.5
U3 - Beta(0.5, 0.5)	12.9	11.7	25.3	38.6	6.9	4.6
U3 - Beta(0.7, 0.3)	13.6	12.5	24.3	39.5	5.6	4.5
U3 - Beta(1, 1)	14.8	11.6	24.6	37.9	6.8	4.3
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	4.8	7.5	7.9	4.5	-
U2 - Beta(0.5, 0.5)	4.6	4.8	7.5	7.8	4.5	-
U2 - Beta(0.7, 0.3)	4.7	4.8	7.4	8	4.4	-
U2 - Beta(1, 1)	4.7	4.9	7.4	7.8	4.5	-
U3 - Beta(0.3, 0.7)	4.6	4.8	7.5	7.8	4.5	-
U3 - Beta(0.5, 0.5)	4.6	4.8	7.5	7.9	4.4	-
U3 - Beta(0.7, 0.3)	4.7	4.9	7.4	7.9	4.4	-
U3 - Beta(1, 1)	4.7	4.8	7.4	7.8	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	3	2.4	2.6	1.1	-
U2 - Beta(0.5, 0.5)	0.7	3	2.4	2.6	1.1	-
U2 - Beta(0.7, 0.3)	0.8	3	2.3	2.6	1.1	-
U2 - Beta(1, 1)	0.8	3	2.3	2.6	1.1	-
U3 - Beta(0.3, 0.7)	0.7	3	2.4	2.6	1.1	-
U3 - Beta(0.5, 0.5)	0.7	3	2.4	2.6	1.1	-
U3 - Beta(0.7, 0.3)	0.8	3	2.3	2.6	1.1	-
U3 - Beta(1, 1)	0.8	3	2.3	2.6	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1067.8
U2 - Beta(0.5, 0.5)						1067.1
U2 - Beta(0.7, 0.3)						1065.1
U2 - Beta(1, 1)						1066
U3 - Beta(0.3, 0.7)						1068.3
U3 - Beta(0.5, 0.5)						1065.5
U3 - Beta(0.7, 0.3)						1065.8
U3 - Beta(1, 1)						1065.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.3	31.7	28.5	20.1	8.3	8.1
U2 - Beta(0.5, 0.5)	3.6	35.1	26.6	19.6	7	8.1
U2 - Beta(0.7, 0.3)	4.1	36.2	27	18.8	5.8	8.1
U2 - Beta(1, 1)	4.8	36	26.8	18.8	5.5	8.1
U3 - Beta(0.3, 0.7)	5.2	36.8	26.6	17.5	5.8	8.1
U3 - Beta(0.5, 0.5)	6.1	40.8	24.2	16.1	4.7	8.1
U3 - Beta(0.7, 0.3)	6.9	41.6	24	15.6	3.8	8.1
U3 - Beta(1, 1)	6.8	41.9	23.1	16.4	3.7	8.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.8	6.6	6.9	5.8	4.4	-
U2 - Beta(0.5, 0.5)	4.8	6.8	6.9	5.8	4.3	-
U2 - Beta(0.7, 0.3)	4.8	6.9	6.8	5.8	4.2	-
U2 - Beta(1, 1)	4.8	6.9	6.7	5.8	4.3	-
U3 - Beta(0.3, 0.7)	4.8	6.5	7	5.7	4.3	-
U3 - Beta(0.5, 0.5)	4.8	6.8	6.9	5.8	4.3	-
U3 - Beta(0.7, 0.3)	4.8	6.9	6.8	5.9	4.1	-
U3 - Beta(1, 1)	4.8	6.9	6.8	5.8	4.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	3.5	2.2	2.2	1.6	-
U2 - Beta(0.5, 0.5)	0.8	3.6	2.2	2.2	1.5	-
U2 - Beta(0.7, 0.3)	0.8	3.7	2.1	2.2	1.5	-
U2 - Beta(1, 1)	0.8	3.6	2.1	2.2	1.5	-
U3 - Beta(0.3, 0.7)	0.8	3.5	2.2	2.2	1.5	-
U3 - Beta(0.5, 0.5)	0.8	3.6	2.2	2.2	1.6	-
U3 - Beta(0.7, 0.3)	0.8	3.6	2.1	2.2	1.5	-
U3 - Beta(1, 1)	0.8	3.6	2.1	2.2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1001.4
U2 - Beta(0.5, 0.5)						1000.9
U2 - Beta(0.7, 0.3)						1000.3
U2 - Beta(1, 1)						1000.6
U3 - Beta(0.3, 0.7)						1000.7
U3 - Beta(0.5, 0.5)						1001
U3 - Beta(0.7, 0.3)						1000.7
U3 - Beta(1, 1)						1000.7
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	65.4	6.2	1.2	0.2	0	27
U2 - Beta(0.5, 0.5)	64.6	6.5	1.3	0.1	0	27.5
U2 - Beta(0.7, 0.3)	64.9	7	0.9	0.2	0	27
U2 - Beta(1, 1)	64.5	6.7	1.1	0.3	0	27.4
U3 - Beta(0.3, 0.7)	66.4	5.4	0.9	0.1	0	27.2
U3 - Beta(0.5, 0.5)	66	6	0.8	0.2	0	27
U3 - Beta(0.7, 0.3)	65.5	6	0.8	0.1	0	27.6
U3 - Beta(1, 1)	65.9	6.2	0.7	0.2	0	27
Mean administrations						
U2 - Beta(0.3, 0.7)	20.5	3.1	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	20.5	3.2	1.4	0.3	0	-
U2 - Beta(0.7, 0.3)	20.4	3.2	1.3	0.3	0	-
U2 - Beta(1, 1)	20.5	3.2	1.4	0.3	0	-
U3 - Beta(0.3, 0.7)	20.4	3.2	1.4	0.3	0	-
U3 - Beta(0.5, 0.5)	20.5	3.2	1.4	0.3	0	-
U3 - Beta(0.7, 0.3)	20.4	3.2	1.3	0.3	0	-
U3 - Beta(1, 1)	20.4	3.3	1.3	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.8	1.3	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	5.7	1.3	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	5.7	1.3	0.4	0.1	0	-
U2 - Beta(1, 1)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(1, 1)	5.7	1.3	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						979.6
U2 - Beta(0.5, 0.5)						978.2
U2 - Beta(0.7, 0.3)						979.6
U2 - Beta(1, 1)						978.3
U3 - Beta(0.3, 0.7)						978.6
U3 - Beta(0.5, 0.5)						979.3
U3 - Beta(0.7, 0.3)						977.5
U3 - Beta(1, 1)						978.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	35	38.3	16	5.6	1.5	3.6
U2 - Beta(0.5, 0.5)	35.4	39.2	15.9	4.8	1.3	3.4
U2 - Beta(0.7, 0.3)	34.3	39.1	16	6	0.9	3.7
U2 - Beta(1, 1)	34.1	39.2	16.2	5.6	1.4	3.5
U3 - Beta(0.3, 0.7)	35.6	39.9	14.8	5.1	1.1	3.5
U3 - Beta(0.5, 0.5)	36	41.6	13.4	4.8	0.7	3.5
U3 - Beta(0.7, 0.3)	35.3	41.8	13.9	5	0.5	3.5
U3 - Beta(1, 1)	35.5	41.3	14	4.6	0.9	3.7
Mean administrations						
U2 - Beta(0.3, 0.7)	13	7.3	5.1	2.7	1.2	-
U2 - Beta(0.5, 0.5)	13	7.5	4.9	2.7	1.3	-
U2 - Beta(0.7, 0.3)	13	7.4	5	2.6	1.2	-
U2 - Beta(1, 1)	13	7.4	5	2.7	1.3	-
U3 - Beta(0.3, 0.7)	13	7.3	5.1	2.7	1.2	-
U3 - Beta(0.5, 0.5)	13	7.5	5.1	2.6	1.2	-
U3 - Beta(0.7, 0.3)	13	7.5	5	2.7	1.2	-
U3 - Beta(1, 1)	13	7.3	5.1	2.7	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	3	1.3	0.8	0.3	-
U2 - Beta(0.5, 0.5)	3.8	3	1.2	0.8	0.4	-
U2 - Beta(0.7, 0.3)	3.8	3	1.2	0.8	0.4	-
U2 - Beta(1, 1)	3.8	3	1.3	0.8	0.4	-
U3 - Beta(0.3, 0.7)	3.8	3	1.3	0.8	0.4	-
U3 - Beta(0.5, 0.5)	3.8	3	1.3	0.8	0.4	-
U3 - Beta(0.7, 0.3)	3.8	3	1.2	0.8	0.4	-
U3 - Beta(1, 1)	3.8	2.9	1.3	0.8	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.5
U2 - Beta(0.5, 0.5)						1110.4
U2 - Beta(0.7, 0.3)						1109.1
U2 - Beta(1, 1)						1109.8
U3 - Beta(0.3, 0.7)						1109.7
U3 - Beta(0.5, 0.5)						1110.2
U3 - Beta(0.7, 0.3)						1109.2
U3 - Beta(1, 1)						1109
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.8	2.9	0.5	0	0	69.8
U2 - Beta(0.5, 0.5)	26.9	2.3	0.6	0	0	70.2
U2 - Beta(0.7, 0.3)	26.5	3	0.5	0	0	70
U2 - Beta(1, 1)	27.3	2.4	0.6	0	0	69.7
U3 - Beta(0.3, 0.7)	27	2.6	0.4	0	0	70
U3 - Beta(0.5, 0.5)	27.2	2.3	0.4	0	0	70.1
U3 - Beta(0.7, 0.3)	27.5	2.4	0.2	0	0	69.9
U3 - Beta(1, 1)	28	2.1	0.2	0	0	69.7
Mean administrations						
U2 - Beta(0.3, 0.7)	14.8	2.3	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	14.9	2.3	0.8	0.1	0	-
U2 - Beta(0.7, 0.3)	14.9	2.3	0.8	0.1	0	-
U2 - Beta(1, 1)	15	2.3	0.9	0.1	0	-
U3 - Beta(0.3, 0.7)	14.8	2.3	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	14.9	2.3	0.8	0.1	0	-
U3 - Beta(0.7, 0.3)	14.9	2.3	0.8	0.1	0	-
U3 - Beta(1, 1)	14.9	2.2	0.9	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.5	1.2	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	4.6	1.2	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	4.6	1.2	0.3	0.1	0	-
U2 - Beta(1, 1)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	4.5	1.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(1, 1)	4.6	1.2	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						655.3
U2 - Beta(0.5, 0.5)						656.6
U2 - Beta(0.7, 0.3)						655.8
U2 - Beta(1, 1)						658.4
U3 - Beta(0.3, 0.7)						654.9
U3 - Beta(0.5, 0.5)						656.6
U3 - Beta(0.7, 0.3)						655.8
U3 - Beta(1, 1)						656.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.5	14.1	17.5	10.2	11.2	35.5
U2 - Beta(0.5, 0.5)	12.5	13.9	17.2	9.9	11.2	35.3
U2 - Beta(0.7, 0.3)	12.9	13.7	17.6	10.4	10.2	35.2
U2 - Beta(1, 1)	13.7	14.1	16.4	9.6	10.9	35.3
U3 - Beta(0.3, 0.7)	15.2	14.1	16	9.2	10.3	35.2
U3 - Beta(0.5, 0.5)	16.8	13.3	16.2	9	9.3	35.4
U3 - Beta(0.7, 0.3)	19.3	13.6	13.7	9	9.1	35.3
U3 - Beta(1, 1)	20.1	13.8	12.4	9.3	9.1	35.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	4.9	5.2	3.1	2.6	-
U2 - Beta(0.5, 0.5)	8.5	5	5.1	3.1	2.6	-
U2 - Beta(0.7, 0.3)	8.5	5.1	5	3.1	2.6	-
U2 - Beta(1, 1)	8.6	5	5	3.1	2.6	-
U3 - Beta(0.3, 0.7)	8.4	5	5.2	3	2.7	-
U3 - Beta(0.5, 0.5)	8.5	4.9	5.1	3.1	2.6	-
U3 - Beta(0.7, 0.3)	8.5	5	5	3.1	2.6	-
U3 - Beta(1, 1)	8.5	5.1	4.9	3.2	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	2.8	1.9	1.3	0.8	-
U2 - Beta(0.5, 0.5)	2.3	2.8	1.8	1.3	0.8	-
U2 - Beta(0.7, 0.3)	2.3	2.9	1.8	1.3	0.8	-
U2 - Beta(1, 1)	2.4	2.8	1.8	1.3	0.8	-
U3 - Beta(0.3, 0.7)	2.3	2.8	1.9	1.2	0.8	-
U3 - Beta(0.5, 0.5)	2.3	2.8	1.8	1.3	0.8	-
U3 - Beta(0.7, 0.3)	2.3	2.8	1.8	1.3	0.8	-
U3 - Beta(1, 1)	2.3	2.8	1.7	1.3	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						836.8
U2 - Beta(0.5, 0.5)						837.8
U2 - Beta(0.7, 0.3)						837.5
U2 - Beta(1, 1)						837.8
U3 - Beta(0.3, 0.7)						838.1
U3 - Beta(0.5, 0.5)						837.2
U3 - Beta(0.7, 0.3)						837.7
U3 - Beta(1, 1)						837.1
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.4	20.2	6.9	1.6	0.5	59.4
U2 - Beta(0.5, 0.5)	11	20.9	6.7	1.5	0.4	59.5
U2 - Beta(0.7, 0.3)	10.4	22.4	5.6	1.7	0.3	59.6
U2 - Beta(1, 1)	10.7	21.4	6.3	1.7	0.4	59.5
U3 - Beta(0.3, 0.7)	11.5	20.4	6.5	1.5	0.5	59.6
U3 - Beta(0.5, 0.5)	11.6	21.2	5.8	1.6	0.3	59.5
U3 - Beta(0.7, 0.3)	11.1	23	4.7	1.5	0.2	59.5
U3 - Beta(1, 1)	11.5	21.5	5.5	1.7	0.3	59.5
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	5.1	2.9	0.9	0.3	-
U2 - Beta(0.5, 0.5)	10.4	5.1	2.9	1	0.3	-
U2 - Beta(0.7, 0.3)	10.4	5.3	2.8	1	0.2	-
U2 - Beta(1, 1)	10.4	5.2	2.8	1	0.3	-
U3 - Beta(0.3, 0.7)	10.4	5	2.9	1	0.3	-
U3 - Beta(0.5, 0.5)	10.3	5.2	2.9	1	0.2	-
U3 - Beta(0.7, 0.3)	10.4	5.2	2.8	0.9	0.3	-
U3 - Beta(1, 1)	10.4	5.2	2.8	1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.3	0.9	0.3	0.1	-
U2 - Beta(0.5, 0.5)	3	2.3	0.8	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3	2.4	0.8	0.3	0.1	-
U2 - Beta(1, 1)	3	2.3	0.8	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3	2.3	0.9	0.3	0.1	-
U3 - Beta(0.5, 0.5)	2.9	2.3	0.8	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3	2.3	0.8	0.3	0.1	-
U3 - Beta(1, 1)	3	2.3	0.8	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						714.1
U2 - Beta(0.5, 0.5)						714.5
U2 - Beta(0.7, 0.3)						714.7
U2 - Beta(1, 1)						714.8
U3 - Beta(0.3, 0.7)						714.7
U3 - Beta(0.5, 0.5)						714.8
U3 - Beta(0.7, 0.3)						714.5
U3 - Beta(1, 1)						714

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	73	12.5	0.6	0.1	0	13.8
U2 - Beta(0.5, 0.5)	72.1	13.4	0.5	0.1	0	13.9
U2 - Beta(0.7, 0.3)	71.6	13.7	0.7	0.1	0	13.9
U2 - Beta(1, 1)	71.7	13.7	0.7	0.1	0	13.8
U3 - Beta(0.3, 0.7)	74.5	11.5	0.5	0	0	13.5
U3 - Beta(0.5, 0.5)	73	12.6	0.5	0.1	0	13.8
U3 - Beta(0.7, 0.3)	72.5	13.3	0.6	0	0	13.6
U3 - Beta(1, 1)	72.6	13.3	0.5	0	0	13.6
Mean administrations						
U2 - Beta(0.3, 0.7)	20.3	5.2	1.9	0.3	0	-
U2 - Beta(0.5, 0.5)	20.3	5.2	1.9	0.3	0	-
U2 - Beta(0.7, 0.3)	20.3	5.2	1.9	0.3	0	-
U2 - Beta(1, 1)	20.3	5.2	1.9	0.3	0	-
U3 - Beta(0.3, 0.7)	20.3	5.2	2	0.3	0	-
U3 - Beta(0.5, 0.5)	20.3	5.2	1.9	0.3	0	-
U3 - Beta(0.7, 0.3)	20.3	5.2	1.9	0.3	0	-
U3 - Beta(1, 1)	20.4	5.2	1.8	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	2.4	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	6.4	2.4	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	6.4	2.4	0.6	0.1	0	-
U2 - Beta(1, 1)	6.5	2.4	0.6	0.1	0	-
U3 - Beta(0.3, 0.7)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	6.5	2.4	0.6	0.1	0	-
U3 - Beta(1, 1)	6.5	2.4	0.6	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						995.7
U2 - Beta(0.5, 0.5)						995.1
U2 - Beta(0.7, 0.3)						994.9
U2 - Beta(1, 1)						995.6
U3 - Beta(0.3, 0.7)						996.2
U3 - Beta(0.5, 0.5)						995.5
U3 - Beta(0.7, 0.3)						995.2
U3 - Beta(1, 1)						995.7
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.7	19.3	15.1	35.8	9.2	6.9
U2 - Beta(0.5, 0.5)	16.3	17.9	15.5	34	9.9	6.4
U2 - Beta(0.7, 0.3)	16.2	19.4	13.9	34.6	9.2	6.7
U2 - Beta(1, 1)	16	19.5	14.4	34.1	9.2	6.8
U3 - Beta(0.3, 0.7)	18.8	20.1	12.4	33.2	8.8	6.7
U3 - Beta(0.5, 0.5)	21.5	18.7	10.7	33.5	9.1	6.5
U3 - Beta(0.7, 0.3)	23.2	19.7	9.5	32.9	8	6.7
U3 - Beta(1, 1)	24.8	19	8	32.1	9.5	6.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	5.4	6.2	7.4	4.8	-
U2 - Beta(0.5, 0.5)	5.2	5.5	6.3	7.3	4.7	-
U2 - Beta(0.7, 0.3)	5.3	5.6	6.2	7.2	4.6	-
U2 - Beta(1, 1)	5.3	5.6	6.1	7.2	4.7	-
U3 - Beta(0.3, 0.7)	5.1	5.4	6.4	7.3	4.7	-
U3 - Beta(0.5, 0.5)	5.2	5.5	6.3	7.2	4.7	-
U3 - Beta(0.7, 0.3)	5.3	5.6	6.1	7.3	4.6	-
U3 - Beta(1, 1)	5.4	5.6	6	7.1	4.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.3	2	2.8	1.1	-
U2 - Beta(0.5, 0.5)	1	3.3	2.1	2.7	1.1	-
U2 - Beta(0.7, 0.3)	1.1	3.3	2	2.7	1.1	-
U2 - Beta(1, 1)	1.1	3.3	2	2.7	1.1	-
U3 - Beta(0.3, 0.7)	1	3.2	2.1	2.7	1.1	-
U3 - Beta(0.5, 0.5)	1	3.3	2.1	2.7	1.1	-
U3 - Beta(0.7, 0.3)	1.1	3.3	2	2.7	1.1	-
U3 - Beta(1, 1)	1.1	3.3	2	2.7	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.7
U2 - Beta(0.5, 0.5)						1030.6
U2 - Beta(0.7, 0.3)						1030.2
U2 - Beta(1, 1)						1030.4
U3 - Beta(0.3, 0.7)						1032.8
U3 - Beta(0.5, 0.5)						1030.9
U3 - Beta(0.7, 0.3)						1032.1
U3 - Beta(1, 1)						1031.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.3	22.6	7.5	1.5	0.3	54.8
U2 - Beta(0.5, 0.5)	13.6	23.6	6.4	1.2	0.3	54.9
U2 - Beta(0.7, 0.3)	13.2	23.9	6.7	1.7	0.1	54.4
U2 - Beta(1, 1)	13.1	24.4	6.3	1.4	0.3	54.5
U3 - Beta(0.3, 0.7)	14.3	23.3	6.5	0.9	0.3	54.7
U3 - Beta(0.5, 0.5)	14.8	23.5	6.1	1	0.3	54.3
U3 - Beta(0.7, 0.3)	13.6	25	5.6	1	0.2	54.6
U3 - Beta(1, 1)	15	23.4	5.9	1.1	0.3	54.3
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	5.5	3.2	1.1	0.3	-
U2 - Beta(0.5, 0.5)	10.4	5.7	3.1	1.1	0.3	-
U2 - Beta(0.7, 0.3)	10.3	5.7	3.1	1.1	0.3	-
U2 - Beta(1, 1)	10.4	5.6	3.1	1.1	0.3	-
U3 - Beta(0.3, 0.7)	10.3	5.6	3.2	1.1	0.3	-
U3 - Beta(0.5, 0.5)	10.3	5.6	3.1	1.1	0.3	-
U3 - Beta(0.7, 0.3)	10.3	5.7	3.1	1	0.3	-
U3 - Beta(1, 1)	10.4	5.6	3.1	1.1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.6	1	0.4	0.1	-
U2 - Beta(0.5, 0.5)	3	2.7	1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3	2.7	1	0.4	0.1	-
U2 - Beta(1, 1)	3	2.7	1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	2.9	2.7	1	0.4	0.1	-
U3 - Beta(0.5, 0.5)	3	2.7	1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3	2.7	1	0.4	0.1	-
U3 - Beta(1, 1)	3	2.6	1	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						732.3
U2 - Beta(0.5, 0.5)						733
U2 - Beta(0.7, 0.3)						731.2
U2 - Beta(1, 1)						730.7
U3 - Beta(0.3, 0.7)						731.6
U3 - Beta(0.5, 0.5)						730.9
U3 - Beta(0.7, 0.3)						731.9
U3 - Beta(1, 1)						732.2
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.1	20.5	18.8	19.3	6.3	18
U2 - Beta(0.5, 0.5)	20.1	20.3	16.9	18.9	6	17.8
U2 - Beta(0.7, 0.3)	20.2	22.1	14.7	19.4	5.7	17.9
U2 - Beta(1, 1)	21.9	20.6	14.5	19.7	5.3	18
U3 - Beta(0.3, 0.7)	24.6	20.7	13.2	17.9	5.6	18
U3 - Beta(0.5, 0.5)	28	20.2	11.3	17.7	4.9	17.9
U3 - Beta(0.7, 0.3)	28.7	20.2	11.3	16.8	4.9	18.1
U3 - Beta(1, 1)	32.5	19.5	8.9	16.9	4.5	17.7
Mean administrations						
U2 - Beta(0.3, 0.7)	6.8	6	6.3	5	3.1	-
U2 - Beta(0.5, 0.5)	7	6	6.3	5	3	-
U2 - Beta(0.7, 0.3)	7.2	6.1	6.1	5	2.9	-
U2 - Beta(1, 1)	7.2	6.2	6	5	2.9	-
U3 - Beta(0.3, 0.7)	7	5.9	6.3	4.9	3.1	-
U3 - Beta(0.5, 0.5)	7	6	6.3	5	3	-
U3 - Beta(0.7, 0.3)	7.1	6.2	6.1	4.9	2.9	-
U3 - Beta(1, 1)	7.1	6.2	6	4.9	3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	3.4	2.2	2	1	-
U2 - Beta(0.5, 0.5)	1.8	3.4	2.2	2	0.9	-
U2 - Beta(0.7, 0.3)	1.8	3.5	2.1	2	0.9	-
U2 - Beta(1, 1)	1.8	3.5	2.1	2	0.9	-
U3 - Beta(0.3, 0.7)	1.7	3.4	2.2	2	0.9	-
U3 - Beta(0.5, 0.5)	1.8	3.4	2.2	2	0.9	-
U3 - Beta(0.7, 0.3)	1.8	3.5	2.1	1.9	0.9	-
U3 - Beta(1, 1)	1.8	3.5	2.1	1.9	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						932.8
U2 - Beta(0.5, 0.5)						930.3
U2 - Beta(0.7, 0.3)						930.1
U2 - Beta(1, 1)						928.9
U3 - Beta(0.3, 0.7)						930.7
U3 - Beta(0.5, 0.5)						931.3
U3 - Beta(0.7, 0.3)						929.5
U3 - Beta(1, 1)						929.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	42.6	29.8	7.4	1.8	0	18.4
U2 - Beta(0.5, 0.5)	44.9	26.9	7.9	1.6	0.1	18.6
U2 - Beta(0.7, 0.3)	44.4	27.7	7.5	2.1	0	18.3
U2 - Beta(1, 1)	47.2	26.8	6.3	1.4	0	18.3
U3 - Beta(0.3, 0.7)	48	27.1	5.5	1	0	18.4
U3 - Beta(0.5, 0.5)	50.8	24.9	4.4	1.4	0.1	18.4
U3 - Beta(0.7, 0.3)	51.1	25.2	4.4	1	0	18.3
U3 - Beta(1, 1)	52.9	24	3.7	1	0	18.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.9	7.6	5.2	2	0.6	-
U2 - Beta(0.5, 0.5)	12	7.6	5.1	1.9	0.6	-
U2 - Beta(0.7, 0.3)	12.2	7.6	5	1.9	0.6	-
U2 - Beta(1, 1)	12.2	7.6	5	1.9	0.6	-
U3 - Beta(0.3, 0.7)	11.9	7.6	5.2	1.9	0.6	-
U3 - Beta(0.5, 0.5)	12.1	7.6	5.1	2	0.6	-
U3 - Beta(0.7, 0.3)	12.1	7.7	5	1.9	0.6	-
U3 - Beta(1, 1)	12.2	7.7	4.9	1.9	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	3.8	1.9	0.9	0.2	-
U2 - Beta(0.5, 0.5)	3.8	3.8	1.8	0.9	0.2	-
U2 - Beta(0.7, 0.3)	3.9	3.8	1.8	0.9	0.2	-
U2 - Beta(1, 1)	3.8	3.8	1.8	0.9	0.2	-
U3 - Beta(0.3, 0.7)	3.8	3.8	1.9	0.9	0.2	-
U3 - Beta(0.5, 0.5)	3.8	3.8	1.8	0.9	0.2	-
U3 - Beta(0.7, 0.3)	3.8	3.8	1.8	0.9	0.2	-
U3 - Beta(1, 1)	3.8	3.9	1.8	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						916
U2 - Beta(0.5, 0.5)						915.8
U2 - Beta(0.7, 0.3)						914.8
U2 - Beta(1, 1)						915.2
U3 - Beta(0.3, 0.7)						915.5
U3 - Beta(0.5, 0.5)						915.5
U3 - Beta(0.7, 0.3)						915
U3 - Beta(1, 1)						916.1

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.2	31.4	17.8	15.2	10.4	0
U2 - Beta(0.5, 0.5)	23.9	33	18.5	14.4	10.2	0
U2 - Beta(0.7, 0.3)	26.3	31.4	17.1	15.3	9.9	0
U2 - Beta(1, 1)	25.9	32.8	16.5	15.1	9.7	0
U3 - Beta(0.3, 0.7)	30.8	35.2	15.9	11.5	6.6	0
U3 - Beta(0.5, 0.5)	31.8	36	14.9	10.9	6.4	0
U3 - Beta(0.7, 0.3)	33.5	36.4	13.9	11.1	5.1	0
U3 - Beta(1, 1)	34.7	36.4	13.6	11.1	4.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	5.9	5.5	5.9	7	-
U2 - Beta(0.5, 0.5)	5.6	6.2	5.5	5.9	6.8	-
U2 - Beta(0.7, 0.3)	5.5	6.2	5.5	6.1	6.7	-
U2 - Beta(1, 1)	5.5	6.2	5.5	6	6.8	-
U3 - Beta(0.3, 0.7)	5.6	5.9	5.5	5.8	7.1	-
U3 - Beta(0.5, 0.5)	5.6	6.1	5.5	6	6.8	-
U3 - Beta(0.7, 0.3)	5.5	6.1	5.5	6.1	6.7	-
U3 - Beta(1, 1)	5.6	6.2	5.5	6	6.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.3	1.9	2.6	2.7	-
U2 - Beta(0.5, 0.5)	1.1	3.4	1.9	2.6	2.7	-
U2 - Beta(0.7, 0.3)	1.1	3.4	1.9	2.7	2.6	-
U2 - Beta(1, 1)	1.1	3.4	1.9	2.7	2.6	-
U3 - Beta(0.3, 0.7)	1.1	3.3	1.9	2.6	2.7	-
U3 - Beta(0.5, 0.5)	1.1	3.4	1.9	2.7	2.6	-
U3 - Beta(0.7, 0.3)	1.1	3.4	1.9	2.7	2.6	-
U3 - Beta(1, 1)	1.1	3.4	1.9	2.7	2.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008
U2 - Beta(0.5, 0.5)						1007.3
U2 - Beta(0.7, 0.3)						1006.8
U2 - Beta(1, 1)						1007.3
U3 - Beta(0.3, 0.7)						1007.5
U3 - Beta(0.5, 0.5)						1007.7
U3 - Beta(0.7, 0.3)						1006.9
U3 - Beta(1, 1)						1007.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	35.7	53.8	9.8	0.3	0	0.4
U2 - Beta(0.5, 0.5)	34	55.2	10	0.3	0	0.5
U2 - Beta(0.7, 0.3)	33.7	56.7	8.9	0.3	0	0.4
U2 - Beta(1, 1)	34.2	55.1	10	0.3	0	0.4
U3 - Beta(0.3, 0.7)	38.2	53.3	8	0.1	0	0.4
U3 - Beta(0.5, 0.5)	36.6	54.9	8	0.1	0	0.4
U3 - Beta(0.7, 0.3)	36	56	7.3	0.3	0	0.4
U3 - Beta(1, 1)	37.2	55.3	7	0.1	0	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	10.6	6.2	1.2	0.2	-
U2 - Beta(0.5, 0.5)	11.7	10.6	6.2	1.2	0.1	-
U2 - Beta(0.7, 0.3)	11.8	10.9	5.9	1.2	0.2	-
U2 - Beta(1, 1)	11.8	10.7	6	1.2	0.1	-
U3 - Beta(0.3, 0.7)	11.7	10.6	6.2	1.2	0.2	-
U3 - Beta(0.5, 0.5)	11.7	10.7	6.1	1.2	0.1	-
U3 - Beta(0.7, 0.3)	11.8	10.8	5.9	1.2	0.1	-
U3 - Beta(1, 1)	11.7	10.7	6.1	1.2	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	3.7	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	3.1	3.6	1.4	0.4	0	-
U2 - Beta(0.7, 0.3)	3.1	3.7	1.4	0.4	0	-
U2 - Beta(1, 1)	3.1	3.7	1.4	0.4	0	-
U3 - Beta(0.3, 0.7)	3.1	3.6	1.4	0.4	0	-
U3 - Beta(0.5, 0.5)	3.1	3.7	1.4	0.4	0	-
U3 - Beta(0.7, 0.3)	3.1	3.7	1.4	0.4	0	-
U3 - Beta(1, 1)	3.1	3.7	1.4	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1177.5
U2 - Beta(0.5, 0.5)						1177.3
U2 - Beta(0.7, 0.3)						1177.9
U2 - Beta(1, 1)						1177
U3 - Beta(0.3, 0.7)						1178
U3 - Beta(0.5, 0.5)						1177.8
U3 - Beta(0.7, 0.3)						1177.5
U3 - Beta(1, 1)						1178
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	13.2	13.2	21.3	43.3	0.9
U2 - Beta(0.5, 0.5)	9	14.3	12.5	19.9	43.4	0.9
U2 - Beta(0.7, 0.3)	9.2	14.6	12	20.3	43	0.9
U2 - Beta(1, 1)	11	14.1	12.5	18.6	42.9	0.9
U3 - Beta(0.3, 0.7)	12.9	15	12.4	19	39.8	0.9
U3 - Beta(0.5, 0.5)	14	15.9	13.4	17.5	38.3	0.9
U3 - Beta(0.7, 0.3)	16	15.9	11.2	17.1	38.9	0.9
U3 - Beta(1, 1)	18	17.4	10.1	15.8	37.8	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	4.1	4.9	6	11.3	-
U2 - Beta(0.5, 0.5)	3.7	4.1	5	6	11.1	-
U2 - Beta(0.7, 0.3)	3.7	4.2	4.9	6.1	10.9	-
U2 - Beta(1, 1)	3.8	4.2	4.9	6	11	-
U3 - Beta(0.3, 0.7)	3.7	4	4.9	6	11.2	-
U3 - Beta(0.5, 0.5)	3.7	4.1	5	6	11	-
U3 - Beta(0.7, 0.3)	3.7	4.3	5	6.1	10.8	-
U3 - Beta(1, 1)	3.7	4.2	5	5.9	11	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.8	1.6	2.6	3.2	-
U2 - Beta(0.5, 0.5)	0.3	2.8	1.6	2.6	3.1	-
U2 - Beta(0.7, 0.3)	0.3	2.9	1.6	2.6	3.1	-
U2 - Beta(1, 1)	0.4	2.9	1.6	2.6	3.1	-
U3 - Beta(0.3, 0.7)	0.3	2.8	1.6	2.7	3.1	-
U3 - Beta(0.5, 0.5)	0.3	2.8	1.7	2.7	3.1	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.6	2.7	3.1	-
U3 - Beta(1, 1)	0.4	2.9	1.6	2.6	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1064.6
U2 - Beta(0.5, 0.5)						1063
U2 - Beta(0.7, 0.3)						1061.5
U2 - Beta(1, 1)						1062.1
U3 - Beta(0.3, 0.7)						1063.9
U3 - Beta(0.5, 0.5)						1062.5
U3 - Beta(0.7, 0.3)						1060.6
U3 - Beta(1, 1)						1062.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.1	48.8	7.2	0.3	0	18.6
U2 - Beta(0.5, 0.5)	25.1	49.1	7.2	0.3	0	18.3
U2 - Beta(0.7, 0.3)	25	48.7	7.4	0.4	0	18.5
U2 - Beta(1, 1)	24.6	49.2	7.4	0.3	0	18.5
U3 - Beta(0.3, 0.7)	27.3	47.6	6.3	0.2	0	18.6
U3 - Beta(0.5, 0.5)	26.9	47.9	6.3	0.3	0	18.6
U3 - Beta(0.7, 0.3)	26.2	48.8	6.3	0.4	0	18.3
U3 - Beta(1, 1)	28.2	47.4	5.8	0.2	0	18.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	10.4	5.5	1	0.1	-
U2 - Beta(0.5, 0.5)	9.9	10.7	5.3	1	0.1	-
U2 - Beta(0.7, 0.3)	10	10.7	5.2	1	0.1	-
U2 - Beta(1, 1)	10	10.6	5.3	1	0.1	-
U3 - Beta(0.3, 0.7)	9.9	10.5	5.5	1	0.1	-
U3 - Beta(0.5, 0.5)	9.9	10.6	5.3	1	0.1	-
U3 - Beta(0.7, 0.3)	9.9	10.7	5.3	1	0.1	-
U3 - Beta(1, 1)	10	10.6	5.2	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.5	1.6	0.4	0	-
U2 - Beta(0.5, 0.5)	2.8	4.6	1.6	0.4	0	-
U2 - Beta(0.7, 0.3)	2.9	4.6	1.5	0.4	0	-
U2 - Beta(1, 1)	2.9	4.6	1.6	0.4	0	-
U3 - Beta(0.3, 0.7)	2.8	4.6	1.6	0.4	0	-
U3 - Beta(0.5, 0.5)	2.8	4.6	1.6	0.4	0	-
U3 - Beta(0.7, 0.3)	2.8	4.7	1.5	0.4	0	-
U3 - Beta(1, 1)	2.9	4.6	1.5	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						964.2
U2 - Beta(0.5, 0.5)						964.5
U2 - Beta(0.7, 0.3)						963.8
U2 - Beta(1, 1)						964.6
U3 - Beta(0.3, 0.7)						965.8
U3 - Beta(0.5, 0.5)						963.2
U3 - Beta(0.7, 0.3)						963.7
U3 - Beta(1, 1)						963
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.5	49	26.8	4.5	0.1	9.1
U2 - Beta(0.5, 0.5)	10.7	50.6	25	4.4	0	9.3
U2 - Beta(0.7, 0.3)	11.1	50	25.1	4.5	0.1	9.2
U2 - Beta(1, 1)	12.5	50.4	24.3	3.7	0.1	9
U3 - Beta(0.3, 0.7)	13.5	51.7	22.2	3.3	0.1	9.2
U3 - Beta(0.5, 0.5)	14	51.1	22	3.8	0.1	9
U3 - Beta(0.7, 0.3)	14.4	52	20.4	4	0.1	9.1
U3 - Beta(1, 1)	16.8	49.6	20.2	4.1	0.1	9.2
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	9.7	8.6	3.3	1	-
U2 - Beta(0.5, 0.5)	5.8	9.8	8.4	3.3	1	-
U2 - Beta(0.7, 0.3)	5.8	10	8.3	3.3	1	-
U2 - Beta(1, 1)	5.8	9.9	8.3	3.3	1	-
U3 - Beta(0.3, 0.7)	5.7	9.7	8.6	3.3	1	-
U3 - Beta(0.5, 0.5)	5.7	9.8	8.6	3.3	1	-
U3 - Beta(0.7, 0.3)	5.8	10	8.3	3.3	1	-
U3 - Beta(1, 1)	5.8	10	8.3	3.2	1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	4.7	2.7	1.3	0.4	-
U2 - Beta(0.5, 0.5)	1.2	4.7	2.6	1.3	0.4	-
U2 - Beta(0.7, 0.3)	1.2	4.8	2.6	1.3	0.4	-
U2 - Beta(1, 1)	1.2	4.8	2.6	1.3	0.4	-
U3 - Beta(0.3, 0.7)	1.2	4.7	2.7	1.3	0.4	-
U3 - Beta(0.5, 0.5)	1.2	4.8	2.7	1.3	0.4	-
U3 - Beta(0.7, 0.3)	1.2	4.8	2.6	1.3	0.4	-
U3 - Beta(1, 1)	1.2	4.8	2.6	1.3	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.2
U2 - Beta(0.5, 0.5)						993.2
U2 - Beta(0.7, 0.3)						992.6
U2 - Beta(1, 1)						993.4
U3 - Beta(0.3, 0.7)						994.8
U3 - Beta(0.5, 0.5)						994
U3 - Beta(0.7, 0.3)						992.6
U3 - Beta(1, 1)						993.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.7	0.1	0	0	0	62.2
U2 - Beta(0.5, 0.5)	37.4	0.2	0	0	0	62.4
U2 - Beta(0.7, 0.3)	38.2	0.2	0	0	0	61.6
U2 - Beta(1, 1)	37.8	0.1	0	0	0	62.1
U3 - Beta(0.3, 0.7)	38.2	0	0	0	0	61.8
U3 - Beta(0.5, 0.5)	37.8	0.2	0	0	0	62
U3 - Beta(0.7, 0.3)	37.9	0.2	0	0	0	61.9
U3 - Beta(1, 1)	37.7	0	0	0	0	62.3
Mean administrations						
U2 - Beta(0.3, 0.7)	18.7	1.4	0.2	0	0	-
U2 - Beta(0.5, 0.5)	18.7	1.4	0.2	0	0	-
U2 - Beta(0.7, 0.3)	18.7	1.4	0.2	0	0	-
U2 - Beta(1, 1)	18.7	1.4	0.2	0	0	-
U3 - Beta(0.3, 0.7)	18.7	1.4	0.2	0	0	-
U3 - Beta(0.5, 0.5)	18.8	1.4	0.2	0	0	-
U3 - Beta(0.7, 0.3)	18.7	1.4	0.2	0	0	-
U3 - Beta(1, 1)	18.7	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	0.5	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.8	0.5	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.8	0.5	0.1	0	0	-
U2 - Beta(1, 1)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.8	0.5	0.1	0	0	-
U3 - Beta(1, 1)	3.8	0.5	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						879.8
U2 - Beta(0.5, 0.5)						878.7
U2 - Beta(0.7, 0.3)						879.9
U2 - Beta(1, 1)						881.1
U3 - Beta(0.3, 0.7)						881.7
U3 - Beta(0.5, 0.5)						882.7
U3 - Beta(0.7, 0.3)						879.3
U3 - Beta(1, 1)						880.7
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.7	51.1	5	0.1	0	12.1
U2 - Beta(0.5, 0.5)	30.2	52	5.6	0.1	0	12.1
U2 - Beta(0.7, 0.3)	29.6	51.7	7	0.1	0	11.6
U2 - Beta(1, 1)	31.4	50.4	6.4	0.1	0	11.7
U3 - Beta(0.3, 0.7)	33.7	50.1	4.5	0.1	0	11.6
U3 - Beta(0.5, 0.5)	33	49.8	5.1	0.1	0	12
U3 - Beta(0.7, 0.3)	33	50	5.3	0.1	0	11.6
U3 - Beta(1, 1)	35.9	47.1	5	0.1	0	11.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.5	12.8	4.8	0.7	0	-
U2 - Beta(0.5, 0.5)	9.6	12.8	4.8	0.7	0	-
U2 - Beta(0.7, 0.3)	9.7	12.7	4.7	0.7	0	-
U2 - Beta(1, 1)	9.7	12.8	4.7	0.7	0	-
U3 - Beta(0.3, 0.7)	9.5	12.9	4.8	0.7	0	-
U3 - Beta(0.5, 0.5)	9.5	12.8	4.8	0.7	0	-
U3 - Beta(0.7, 0.3)	9.6	12.8	4.7	0.7	0	-
U3 - Beta(1, 1)	9.7	12.8	4.7	0.7	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	5.6	1.4	0.3	0	-
U2 - Beta(0.5, 0.5)	2.7	5.6	1.4	0.3	0	-
U2 - Beta(0.7, 0.3)	2.7	5.6	1.4	0.3	0	-
U2 - Beta(1, 1)	2.7	5.6	1.4	0.3	0	-
U3 - Beta(0.3, 0.7)	2.6	5.6	1.4	0.3	0	-
U3 - Beta(0.5, 0.5)	2.7	5.6	1.4	0.3	0	-
U3 - Beta(0.7, 0.3)	2.7	5.6	1.4	0.3	0	-
U3 - Beta(1, 1)	2.7	5.6	1.4	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						980.3
U2 - Beta(0.5, 0.5)						980.3
U2 - Beta(0.7, 0.3)						979.4
U2 - Beta(1, 1)						980
U3 - Beta(0.3, 0.7)						981
U3 - Beta(0.5, 0.5)						981
U3 - Beta(0.7, 0.3)						979.8
U3 - Beta(1, 1)						981.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.7	1.1	0	0	0	93.2
U2 - Beta(0.5, 0.5)	5.6	1	0	0	0	93.4
U2 - Beta(0.7, 0.3)	5.7	1	0	0	0	93.3
U2 - Beta(1, 1)	5.6	1	0	0	0	93.4
U3 - Beta(0.3, 0.7)	5.6	1	0	0	0	93.4
U3 - Beta(0.5, 0.5)	5.6	1	0	0	0	93.4
U3 - Beta(0.7, 0.3)	5.7	1	0	0	0	93.3
U3 - Beta(1, 1)	5.7	1	0	0	0	93.3
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	1.4	0.2	0	0	-
U2 - Beta(0.5, 0.5)	11.8	1.4	0.2	0	0	-
U2 - Beta(0.7, 0.3)	11.8	1.4	0.2	0	0	-
U2 - Beta(1, 1)	11.8	1.4	0.2	0	0	-
U3 - Beta(0.3, 0.7)	11.8	1.4	0.2	0	0	-
U3 - Beta(0.5, 0.5)	11.8	1.4	0.2	0	0	-
U3 - Beta(0.7, 0.3)	11.8	1.4	0.2	0	0	-
U3 - Beta(1, 1)	11.8	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	0.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.5	0.9	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.5	0.9	0.1	0	0	-
U2 - Beta(1, 1)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.5	0.9	0.1	0	0	-
U3 - Beta(1, 1)	3.5	0.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						478
U2 - Beta(0.5, 0.5)						477.3
U2 - Beta(0.7, 0.3)						477.3
U2 - Beta(1, 1)						477.9
U3 - Beta(0.3, 0.7)						477.5
U3 - Beta(0.5, 0.5)						477.4
U3 - Beta(0.7, 0.3)						478.4
U3 - Beta(1, 1)						477.5
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	1.3	3.9	45.9	48.5	0.2
U2 - Beta(0.5, 0.5)	0.3	1.7	3.9	46.9	47.1	0.1
U2 - Beta(0.7, 0.3)	0.8	1.8	4.1	48.5	44.7	0.1
U2 - Beta(1, 1)	0.5	1.6	4.2	46.6	47	0.1
U3 - Beta(0.3, 0.7)	1.2	2.2	4.7	48.1	43.7	0.1
U3 - Beta(0.5, 0.5)	1.5	2.2	4.5	48.6	43.1	0.1
U3 - Beta(0.7, 0.3)	1.8	2.8	5.4	51.7	38.2	0.1
U3 - Beta(1, 1)	2	2.6	5.2	48.9	41.1	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	3.1	3.1	3.5	8.1	12.1	-
U2 - Beta(0.5, 0.5)	3.1	3.1	3.5	8.3	11.9	-
U2 - Beta(0.7, 0.3)	3.2	3.1	3.6	8.7	11.5	-
U2 - Beta(1, 1)	3.2	3.1	3.5	8.2	12	-
U3 - Beta(0.3, 0.7)	3.1	3.1	3.5	8.1	12.1	-
U3 - Beta(0.5, 0.5)	3.1	3.1	3.5	8.2	12	-
U3 - Beta(0.7, 0.3)	3.2	3.1	3.6	8.5	11.6	-
U3 - Beta(1, 1)	3.2	3.1	3.5	8.1	12	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.3	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.3	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.1	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.3	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.3	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.1	3.3	3.2	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102.7
U2 - Beta(0.5, 0.5)						1102.4
U2 - Beta(0.7, 0.3)						1100.6
U2 - Beta(1, 1)						1103
U3 - Beta(0.3, 0.7)						1102.9
U3 - Beta(0.5, 0.5)						1102.6
U3 - Beta(0.7, 0.3)						1101.2
U3 - Beta(1, 1)						1102.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	36.9	37.8	21.9	1.6	0	1.8
U2 - Beta(0.5, 0.5)	37.1	37.3	22.1	1.8	0	1.7
U2 - Beta(0.7, 0.3)	36	37.1	23.1	2	0	1.8
U2 - Beta(1, 1)	36.8	36.9	22.9	1.6	0	1.8
U3 - Beta(0.3, 0.7)	39.9	37.1	19.8	1.5	0	1.7
U3 - Beta(0.5, 0.5)	39.9	36.4	20.3	1.6	0	1.8
U3 - Beta(0.7, 0.3)	40.3	36.3	20	1.6	0	1.8
U3 - Beta(1, 1)	41.6	35.2	19.9	1.6	0	1.7
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	9.2	8.2	1.6	0.2	-
U2 - Beta(0.5, 0.5)	10.7	9.3	8	1.6	0.2	-
U2 - Beta(0.7, 0.3)	10.7	9.4	7.8	1.6	0.2	-
U2 - Beta(1, 1)	10.7	9.3	7.9	1.6	0.2	-
U3 - Beta(0.3, 0.7)	10.5	9.2	8.2	1.6	0.2	-
U3 - Beta(0.5, 0.5)	10.6	9.3	7.9	1.6	0.2	-
U3 - Beta(0.7, 0.3)	10.7	9.4	7.9	1.6	0.2	-
U3 - Beta(1, 1)	10.7	9.3	8	1.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	4	2.3	0.6	0	-
U2 - Beta(0.5, 0.5)	3.1	4.1	2.2	0.6	0	-
U2 - Beta(0.7, 0.3)	3.1	4.1	2.2	0.6	0.1	-
U2 - Beta(1, 1)	3.1	4.1	2.2	0.6	0	-
U3 - Beta(0.3, 0.7)	3	4	2.3	0.6	0	-
U3 - Beta(0.5, 0.5)	3.1	4.1	2.2	0.6	0	-
U3 - Beta(0.7, 0.3)	3.1	4.1	2.2	0.6	0	-
U3 - Beta(1, 1)	3.1	4.1	2.2	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1090.5
U2 - Beta(0.5, 0.5)						1090.9
U2 - Beta(0.7, 0.3)						1088.8
U2 - Beta(1, 1)						1090
U3 - Beta(0.3, 0.7)						1091.7
U3 - Beta(0.5, 0.5)						1090
U3 - Beta(0.7, 0.3)						1088.7
U3 - Beta(1, 1)						1090.3
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.7	28.5	18.5	1.5	0	19.8
U2 - Beta(0.5, 0.5)	31.5	29.1	17.7	1.9	0	19.8
U2 - Beta(0.7, 0.3)	31.6	29.9	16.8	1.9	0	19.8
U2 - Beta(1, 1)	32	28.9	17.2	1.9	0	20
U3 - Beta(0.3, 0.7)	32.6	29	16.9	1.5	0	20
U3 - Beta(0.5, 0.5)	34.1	28	16.6	1.5	0	19.8
U3 - Beta(0.7, 0.3)	34.9	27.7	15.7	1.8	0	19.9
U3 - Beta(1, 1)	34.3	28.6	15.4	1.8	0	19.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	7.2	6.4	1.4	0.1	-
U2 - Beta(0.5, 0.5)	11.8	7.2	6.3	1.5	0.1	-
U2 - Beta(0.7, 0.3)	11.8	7.3	6.2	1.5	0.1	-
U2 - Beta(1, 1)	11.9	7.3	6.1	1.5	0.1	-
U3 - Beta(0.3, 0.7)	11.8	7.2	6.4	1.4	0.1	-
U3 - Beta(0.5, 0.5)	11.8	7.3	6.2	1.5	0.1	-
U3 - Beta(0.7, 0.3)	11.8	7.3	6.2	1.5	0.1	-
U3 - Beta(1, 1)	11.9	7.2	6.2	1.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	3.4	2	0.6	0	-
U2 - Beta(0.5, 0.5)	3.6	3.4	1.9	0.6	0	-
U2 - Beta(0.7, 0.3)	3.7	3.5	1.9	0.6	0	-
U2 - Beta(1, 1)	3.7	3.5	1.9	0.6	0	-
U3 - Beta(0.3, 0.7)	3.6	3.4	2	0.5	0	-
U3 - Beta(0.5, 0.5)	3.7	3.4	1.9	0.6	0	-
U3 - Beta(0.7, 0.3)	3.7	3.5	1.9	0.6	0	-
U3 - Beta(1, 1)	3.7	3.4	1.9	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						951.6
U2 - Beta(0.5, 0.5)						951.4
U2 - Beta(0.7, 0.3)						952
U2 - Beta(1, 1)						951.4
U3 - Beta(0.3, 0.7)						951.3
U3 - Beta(0.5, 0.5)						950.7
U3 - Beta(0.7, 0.3)						951.1
U3 - Beta(1, 1)						951.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.3	29.9	33.5	3.3	0	12
U2 - Beta(0.5, 0.5)	21.2	31	32.8	3.3	0	11.7
U2 - Beta(0.7, 0.3)	22.9	29.3	32.7	3	0.1	12
U2 - Beta(1, 1)	22.5	29.7	32.4	3.3	0	12.1
U3 - Beta(0.3, 0.7)	23.1	31.3	30.7	3	0	11.9
U3 - Beta(0.5, 0.5)	24.6	30.1	30.4	3	0	11.9
U3 - Beta(0.7, 0.3)	25.7	30	29.2	3.4	0	11.7
U3 - Beta(1, 1)	26.8	29.5	28.6	3.3	0	11.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	7.3	8.8	2.6	0.5	-
U2 - Beta(0.5, 0.5)	8.9	7.5	8.7	2.6	0.5	-
U2 - Beta(0.7, 0.3)	9	7.3	8.6	2.6	0.5	-
U2 - Beta(1, 1)	8.9	7.4	8.7	2.6	0.5	-
U3 - Beta(0.3, 0.7)	8.9	7.3	8.8	2.6	0.5	-
U3 - Beta(0.5, 0.5)	8.9	7.5	8.7	2.6	0.5	-
U3 - Beta(0.7, 0.3)	8.9	7.4	8.7	2.6	0.5	-
U3 - Beta(1, 1)	8.9	7.5	8.6	2.6	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.5	2.6	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.6	2.5	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.6	2.5	0.8	0.1	-
U2 - Beta(1, 1)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.5	2.6	0.9	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(1, 1)	2.5	3.6	2.5	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021.3
U2 - Beta(0.5, 0.5)						1019.3
U2 - Beta(0.7, 0.3)						1018.2
U2 - Beta(1, 1)						1018.7
U3 - Beta(0.3, 0.7)						1021.1
U3 - Beta(0.5, 0.5)						1019.8
U3 - Beta(0.7, 0.3)						1018.5
U3 - Beta(1, 1)						1018
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.3	21.6	0.6	0	0	57.5
U2 - Beta(0.5, 0.5)	20	22.4	0.5	0	0	57.1
U2 - Beta(0.7, 0.3)	19.7	22.6	0.8	0	0	56.9
U2 - Beta(1, 1)	20	22.4	0.5	0	0	57.1
U3 - Beta(0.3, 0.7)	21	21.6	0.4	0	0	57
U3 - Beta(0.5, 0.5)	20.1	21.7	0.5	0	0	57.7
U3 - Beta(0.7, 0.3)	20.5	22	0.5	0	0	57
U3 - Beta(1, 1)	20.3	22.2	0.5	0	0	57
Mean administrations						
U2 - Beta(0.3, 0.7)	10.9	6.8	2.4	0.3	0	-
U2 - Beta(0.5, 0.5)	10.9	6.9	2.3	0.3	0	-
U2 - Beta(0.7, 0.3)	11	6.9	2.3	0.3	0	-
U2 - Beta(1, 1)	11	6.8	2.3	0.3	0	-
U3 - Beta(0.3, 0.7)	10.9	6.8	2.4	0.3	0	-
U3 - Beta(0.5, 0.5)	10.9	6.9	2.3	0.3	0	-
U3 - Beta(0.7, 0.3)	10.9	6.9	2.4	0.2	0	-
U3 - Beta(1, 1)	10.9	6.9	2.3	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	3.2	3	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	3.2	3	0.7	0.1	0	-
U2 - Beta(1, 1)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	3.2	3	0.7	0.1	0	-
U3 - Beta(1, 1)	3.2	3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						731.3
U2 - Beta(0.5, 0.5)						731.7
U2 - Beta(0.7, 0.3)						732.5
U2 - Beta(1, 1)						731.7
U3 - Beta(0.3, 0.7)						733
U3 - Beta(0.5, 0.5)						730.6
U3 - Beta(0.7, 0.3)						731.9
U3 - Beta(1, 1)						732

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	77.1	18.2	1	0	0	3.7
U2 - Beta(0.5, 0.5)	75.7	19.2	1.4	0	0	3.7
U2 - Beta(0.7, 0.3)	75.4	19.4	1.4	0	0	3.8
U2 - Beta(1, 1)	75.9	19	1.4	0	0	3.7
U3 - Beta(0.3, 0.7)	78.6	16.8	0.9	0	0	3.7
U3 - Beta(0.5, 0.5)	78.2	17.1	1	0	0	3.7
U3 - Beta(0.7, 0.3)	76.1	19.1	1.1	0	0	3.7
U3 - Beta(1, 1)	77.6	18	0.8	0	0	3.6
Mean administrations						
U2 - Beta(0.3, 0.7)	20	6.8	2.3	0.2	0	-
U2 - Beta(0.5, 0.5)	20.1	6.7	2.4	0.2	0	-
U2 - Beta(0.7, 0.3)	20.1	6.6	2.3	0.2	0	-
U2 - Beta(1, 1)	20.1	6.7	2.3	0.2	0	-
U3 - Beta(0.3, 0.7)	19.9	6.8	2.4	0.2	0	-
U3 - Beta(0.5, 0.5)	20.1	6.7	2.3	0.2	0	-
U3 - Beta(0.7, 0.3)	20.2	6.6	2.3	0.2	0	-
U3 - Beta(1, 1)	20.1	6.7	2.3	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.2	2.3	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	5.3	2.3	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	5.3	2.3	0.7	0.1	0	-
U2 - Beta(1, 1)	5.3	2.3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	5.2	2.3	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	5.3	2.3	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	5.3	2.3	0.7	0.1	0	-
U3 - Beta(1, 1)	5.3	2.3	0.6	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1160
U2 - Beta(0.5, 0.5)						1158.7
U2 - Beta(0.7, 0.3)						1158.5
U2 - Beta(1, 1)						1158.3
U3 - Beta(0.3, 0.7)						1160.2
U3 - Beta(0.5, 0.5)						1158.9
U3 - Beta(0.7, 0.3)						1158.3
U3 - Beta(1, 1)						1159.1

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.9	18.7	20.9	36.8	4.1	1.6
U2 - Beta(0.5, 0.5)	17.4	20.3	20.6	35.1	5	1.6
U2 - Beta(0.7, 0.3)	18.3	20.8	20	34.3	5	1.6
U2 - Beta(1, 1)	18.9	20.7	18.8	34.1	5.9	1.6
U3 - Beta(0.3, 0.7)	22.9	22	17.1	33.3	3.1	1.6
U3 - Beta(0.5, 0.5)	24.5	21.4	17.7	31.5	3.3	1.6
U3 - Beta(0.7, 0.3)	25	23.6	15.7	30.5	3.6	1.6
U3 - Beta(1, 1)	26.9	23.2	15.4	29.1	3.8	1.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	5.3	6.8	8.2	4.2	-
U2 - Beta(0.5, 0.5)	5.4	5.3	6.8	8	4.2	-
U2 - Beta(0.7, 0.3)	5.5	5.5	6.7	7.9	4.1	-
U2 - Beta(1, 1)	5.5	5.3	6.7	8	4.2	-
U3 - Beta(0.3, 0.7)	5.4	5.2	6.7	8.2	4.2	-
U3 - Beta(0.5, 0.5)	5.4	5.3	6.8	8	4.2	-
U3 - Beta(0.7, 0.3)	5.4	5.4	6.9	7.9	4.2	-
U3 - Beta(1, 1)	5.5	5.4	6.7	8	4.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.1	2.2	2.9	1.1	-
U2 - Beta(0.5, 0.5)	1.1	3.1	2.2	2.8	1.1	-
U2 - Beta(0.7, 0.3)	1.1	3.2	2.2	2.8	1.1	-
U2 - Beta(1, 1)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(0.3, 0.7)	1.1	3	2.2	2.9	1.1	-
U3 - Beta(0.5, 0.5)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(0.7, 0.3)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(1, 1)	1.1	3.1	2.1	2.8	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1073.2
U2 - Beta(0.5, 0.5)						1073.2
U2 - Beta(0.7, 0.3)						1071.1
U2 - Beta(1, 1)						1071.7
U3 - Beta(0.3, 0.7)						1072.7
U3 - Beta(0.5, 0.5)						1072.2
U3 - Beta(0.7, 0.3)						1070.2
U3 - Beta(1, 1)						1071.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.5	33.5	31.8	26.5	3.6	0.1
U2 - Beta(0.5, 0.5)	4.5	34.5	30.5	25.6	4.8	0.1
U2 - Beta(0.7, 0.3)	4.8	35.9	29.8	23.6	5.8	0.1
U2 - Beta(1, 1)	5.3	34.6	30.2	25.5	4.3	0.1
U3 - Beta(0.3, 0.7)	6.9	38.4	29.8	22	2.8	0.1
U3 - Beta(0.5, 0.5)	6.1	41.2	28.7	20.9	3	0.1
U3 - Beta(0.7, 0.3)	7.1	42.6	26.5	20.4	3.3	0.1
U3 - Beta(1, 1)	6.9	41	28.2	21.9	1.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	5.8	6.7	8.2	5.1	-
U2 - Beta(0.5, 0.5)	4.2	5.9	6.7	8.2	5.1	-
U2 - Beta(0.7, 0.3)	4.2	5.9	6.7	8.1	5	-
U2 - Beta(1, 1)	4.2	5.7	6.7	8.3	5.2	-
U3 - Beta(0.3, 0.7)	4.2	5.7	6.7	8.2	5.1	-
U3 - Beta(0.5, 0.5)	4.2	5.8	6.8	8.1	5.1	-
U3 - Beta(0.7, 0.3)	4.2	5.9	6.7	8.1	5	-
U3 - Beta(1, 1)	4.2	5.8	6.7	8.3	5.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2.1	1.3	-
U2 - Beta(1, 1)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(0.3, 0.7)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(0.5, 0.5)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(1, 1)	0.5	2.9	1.7	2.1	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1182
U2 - Beta(0.5, 0.5)						1180.7
U2 - Beta(0.7, 0.3)						1180.6
U2 - Beta(1, 1)						1181.4
U3 - Beta(0.3, 0.7)						1181.3
U3 - Beta(0.5, 0.5)						1180.9
U3 - Beta(0.7, 0.3)						1180.5
U3 - Beta(1, 1)						1181.9
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.3	13.9	34.9	31.6	9.5	0.8
U2 - Beta(0.5, 0.5)	9.4	13.5	35.1	31.5	9.7	0.8
U2 - Beta(0.7, 0.3)	8.9	14.4	34.7	32.4	8.8	0.8
U2 - Beta(1, 1)	9.8	14.4	34.9	32.7	7.4	0.8
U3 - Beta(0.3, 0.7)	12.7	18.1	34.4	27.7	6.4	0.7
U3 - Beta(0.5, 0.5)	13.1	19.3	34.2	26.9	5.8	0.7
U3 - Beta(0.7, 0.3)	14.3	19.9	33.4	26.7	4.9	0.8
U3 - Beta(1, 1)	15.7	19.2	33	27.2	4.1	0.8
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.3	7.7	7.9	6.1	-
U2 - Beta(0.5, 0.5)	3.8	4.5	7.8	7.9	5.9	-
U2 - Beta(0.7, 0.3)	3.9	4.5	7.9	7.9	5.7	-
U2 - Beta(1, 1)	3.9	4.5	7.7	7.9	5.9	-
U3 - Beta(0.3, 0.7)	3.8	4.3	7.8	7.9	6.1	-
U3 - Beta(0.5, 0.5)	3.8	4.5	7.9	7.9	5.9	-
U3 - Beta(0.7, 0.3)	3.8	4.6	7.8	7.9	5.8	-
U3 - Beta(1, 1)	3.9	4.6	7.7	8	5.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.9	2.4	2.9	2.1	-
U2 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.9	2	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	2	-
U2 - Beta(1, 1)	0.4	2.9	2.5	2.9	2	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.9	2.1	-
U3 - Beta(0.5, 0.5)	0.3	2.9	2.5	2.9	2	-
U3 - Beta(0.7, 0.3)	0.4	3	2.5	2.9	2	-
U3 - Beta(1, 1)	0.4	3	2.4	2.9	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057
U2 - Beta(0.5, 0.5)						1055.8
U2 - Beta(0.7, 0.3)						1055.3
U2 - Beta(1, 1)						1055.6
U3 - Beta(0.3, 0.7)						1058
U3 - Beta(0.5, 0.5)						1057.4
U3 - Beta(0.7, 0.3)						1055.2
U3 - Beta(1, 1)						1056.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.3	9	19.4	44.4	18.8	0.1
U2 - Beta(0.5, 0.5)	8.6	9.7	18.5	45.3	17.8	0.1
U2 - Beta(0.7, 0.3)	9.2	11.1	16.3	46.1	17.1	0.2
U2 - Beta(1, 1)	9.5	11.6	15.9	43.6	19.3	0.1
U3 - Beta(0.3, 0.7)	12.2	11	17.5	42.5	16.7	0.1
U3 - Beta(0.5, 0.5)	13.7	11.7	15.8	43.2	15.5	0.1
U3 - Beta(0.7, 0.3)	14.4	12.9	14.9	41.8	15.9	0.1
U3 - Beta(1, 1)	15.6	12.1	14.5	42.3	15.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.5	3.6	5.7	8.3	9	-
U2 - Beta(0.5, 0.5)	3.5	3.7	5.5	8.4	8.9	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.5	8.4	8.8	-
U2 - Beta(1, 1)	3.5	3.8	5.5	8.3	9	-
U3 - Beta(0.3, 0.7)	3.4	3.6	5.6	8.4	9	-
U3 - Beta(0.5, 0.5)	3.4	3.7	5.6	8.4	8.8	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.6	8.5	8.7	-
U3 - Beta(1, 1)	3.5	3.7	5.5	8.3	8.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.8	3.1	2.3	-
U2 - Beta(0.5, 0.5)	0.2	2.7	1.7	3.1	2.2	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.3	-
U2 - Beta(1, 1)	0.2	2.7	1.7	3.1	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.8	3.1	2.3	-
U3 - Beta(0.5, 0.5)	0.2	2.7	1.8	3.1	2.2	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.2	-
U3 - Beta(1, 1)	0.2	2.7	1.7	3.1	2.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102.4
U2 - Beta(0.5, 0.5)						1101
U2 - Beta(0.7, 0.3)						1099.6
U2 - Beta(1, 1)						1100.7
U3 - Beta(0.3, 0.7)						1102.5
U3 - Beta(0.5, 0.5)						1100.8
U3 - Beta(0.7, 0.3)						1101.3
U3 - Beta(1, 1)						1101.7
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.4	21.3	47	10.2	1.9	6.2
U2 - Beta(0.5, 0.5)	15.4	20.3	44.6	12	1.7	6
U2 - Beta(0.7, 0.3)	17	18.4	46.2	10.4	1.7	6.3
U2 - Beta(1, 1)	16.9	19.6	45.4	9.8	2.1	6.2
U3 - Beta(0.3, 0.7)	19.7	19.6	42.8	9.8	1.6	6.5
U3 - Beta(0.5, 0.5)	21	21	41.1	9.7	1	6.2
U3 - Beta(0.7, 0.3)	22.7	19.6	41.8	8.6	1.3	6
U3 - Beta(1, 1)	24.7	19.1	39.6	9.7	0.9	6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	6.6	10.4	4.5	2	-
U2 - Beta(0.5, 0.5)	5.5	6.7	10.3	4.6	1.9	-
U2 - Beta(0.7, 0.3)	5.6	6.8	10.1	4.5	2	-
U2 - Beta(1, 1)	5.6	6.7	10.1	4.5	2	-
U3 - Beta(0.3, 0.7)	5.4	6.6	10.5	4.5	2	-
U3 - Beta(0.5, 0.5)	5.5	6.7	10.3	4.5	1.9	-
U3 - Beta(0.7, 0.3)	5.6	6.7	10.2	4.5	1.9	-
U3 - Beta(1, 1)	5.7	6.7	10.2	4.5	1.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.6	3.4	1.6	0.6	-
U2 - Beta(0.5, 0.5)	1.2	3.6	3.3	1.6	0.6	-
U2 - Beta(0.7, 0.3)	1.2	3.7	3.3	1.6	0.7	-
U2 - Beta(1, 1)	1.2	3.7	3.3	1.6	0.6	-
U3 - Beta(0.3, 0.7)	1.1	3.6	3.4	1.6	0.6	-
U3 - Beta(0.5, 0.5)	1.2	3.6	3.3	1.6	0.6	-
U3 - Beta(0.7, 0.3)	1.2	3.7	3.3	1.6	0.6	-
U3 - Beta(1, 1)	1.2	3.6	3.3	1.6	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021.9
U2 - Beta(0.5, 0.5)						1022.8
U2 - Beta(0.7, 0.3)						1021.6
U2 - Beta(1, 1)						1020.1
U3 - Beta(0.3, 0.7)						1024
U3 - Beta(0.5, 0.5)						1022.9
U3 - Beta(0.7, 0.3)						1021.1
U3 - Beta(1, 1)						1019.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.7	47.7	33.9	7.3	1.3	0.1
U2 - Beta(0.5, 0.5)	10.2	47.3	32.2	9.4	0.8	0.1
U2 - Beta(0.7, 0.3)	10.2	47.8	31.8	9.3	0.8	0.1
U2 - Beta(1, 1)	10.7	47.4	31.6	9.3	0.9	0.1
U3 - Beta(0.3, 0.7)	11.8	51.3	30.3	5.9	0.6	0.1
U3 - Beta(0.5, 0.5)	12.1	50.4	29.7	7.2	0.5	0.1
U3 - Beta(0.7, 0.3)	13.5	52.2	26.3	7.6	0.3	0.1
U3 - Beta(1, 1)	11.9	53	27.8	6.9	0.3	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	9.5	9.6	4.5	1.8	-
U2 - Beta(0.5, 0.5)	4.6	9.6	9.5	4.5	1.8	-
U2 - Beta(0.7, 0.3)	4.7	9.6	9.4	4.4	1.8	-
U2 - Beta(1, 1)	4.6	9.6	9.4	4.6	1.8	-
U3 - Beta(0.3, 0.7)	4.6	9.5	9.6	4.5	1.9	-
U3 - Beta(0.5, 0.5)	4.7	9.6	9.5	4.4	1.8	-
U3 - Beta(0.7, 0.3)	4.7	9.6	9.4	4.5	1.7	-
U3 - Beta(1, 1)	4.6	9.6	9.5	4.5	1.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	4	2.4	1.4	0.5	-
U2 - Beta(0.5, 0.5)	0.7	4	2.4	1.4	0.5	-
U2 - Beta(0.7, 0.3)	0.7	4	2.4	1.4	0.5	-
U2 - Beta(1, 1)	0.7	4	2.3	1.4	0.5	-
U3 - Beta(0.3, 0.7)	0.6	4	2.4	1.4	0.6	-
U3 - Beta(0.5, 0.5)	0.7	4	2.3	1.4	0.5	-
U3 - Beta(0.7, 0.3)	0.7	4	2.3	1.4	0.5	-
U3 - Beta(1, 1)	0.7	4	2.4	1.4	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1160.5
U2 - Beta(0.5, 0.5)						1161.8
U2 - Beta(0.7, 0.3)						1160.2
U2 - Beta(1, 1)						1162
U3 - Beta(0.3, 0.7)						1160.6
U3 - Beta(0.5, 0.5)						1160.7
U3 - Beta(0.7, 0.3)						1160.5
U3 - Beta(1, 1)						1160.8
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.1	35.2	8.4	1.6	0.5	33.2
U2 - Beta(0.5, 0.5)	21.6	35.5	7.9	1.3	0.5	33.2
U2 - Beta(0.7, 0.3)	21.2	36.1	7.9	1.5	0.3	33
U2 - Beta(1, 1)	21.2	36.1	7.7	1.5	0.5	33
U3 - Beta(0.3, 0.7)	23.8	34.8	7.1	1.1	0.3	32.9
U3 - Beta(0.5, 0.5)	24	34.9	7	1.3	0.3	32.5
U3 - Beta(0.7, 0.3)	24.6	34.1	6.4	1.7	0.2	33
U3 - Beta(1, 1)	25.1	34.1	5.9	1.5	0.4	33
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	8.4	4.6	1.4	0.4	-
U2 - Beta(0.5, 0.5)	9.6	8.4	4.5	1.3	0.4	-
U2 - Beta(0.7, 0.3)	9.6	8.4	4.5	1.4	0.4	-
U2 - Beta(1, 1)	9.7	8.3	4.4	1.4	0.4	-
U3 - Beta(0.3, 0.7)	9.6	8.4	4.5	1.4	0.4	-
U3 - Beta(0.5, 0.5)	9.6	8.4	4.6	1.4	0.4	-
U3 - Beta(0.7, 0.3)	9.6	8.4	4.5	1.4	0.4	-
U3 - Beta(1, 1)	9.7	8.4	4.5	1.4	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	4	1.5	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.7	4	1.5	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.7	4	1.4	0.6	0.1	-
U2 - Beta(1, 1)	2.8	4	1.4	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.7	4	1.5	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.7	4	1.5	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.7	4	1.4	0.6	0.1	-
U3 - Beta(1, 1)	2.7	4	1.4	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						845.5
U2 - Beta(0.5, 0.5)						844.4
U2 - Beta(0.7, 0.3)						844.8
U2 - Beta(1, 1)						845.3
U3 - Beta(0.3, 0.7)						847.3
U3 - Beta(0.5, 0.5)						846.7
U3 - Beta(0.7, 0.3)						846
U3 - Beta(1, 1)						846.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.6	43	6.2	0.3	0	21.9
U2 - Beta(0.5, 0.5)	27.5	43.5	7.4	0.2	0	21.4
U2 - Beta(0.7, 0.3)	27.4	43.2	6.9	0.5	0	22
U2 - Beta(1, 1)	27.8	42.9	6.6	0.6	0	22.1
U3 - Beta(0.3, 0.7)	31.2	42.7	4.8	0.1	0	21.2
U3 - Beta(0.5, 0.5)	30.5	42.3	5.2	0.2	0	21.8
U3 - Beta(0.7, 0.3)	31.3	41.5	5.1	0.3	0	21.8
U3 - Beta(1, 1)	32.7	41.1	4.4	0.3	0	21.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.9	5	1.2	0.2	-
U2 - Beta(0.5, 0.5)	8.9	11	5	1.2	0.2	-
U2 - Beta(0.7, 0.3)	8.9	11.1	4.8	1.2	0.2	-
U2 - Beta(1, 1)	9	11	4.9	1.1	0.2	-
U3 - Beta(0.3, 0.7)	8.8	11	5	1.2	0.2	-
U3 - Beta(0.5, 0.5)	8.9	11	5	1.2	0.2	-
U3 - Beta(0.7, 0.3)	8.9	11.1	4.9	1.1	0.2	-
U3 - Beta(1, 1)	9	11	4.8	1.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	5	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.4	5.1	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.4	5.1	1.5	0.5	0.1	-
U2 - Beta(1, 1)	2.4	5.1	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.4	5.1	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.4	5.1	1.6	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.4	5.1	1.5	0.5	0.1	-
U3 - Beta(1, 1)	2.4	5.1	1.5	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						907.5
U2 - Beta(0.5, 0.5)						907.5
U2 - Beta(0.7, 0.3)						908.5
U2 - Beta(1, 1)						908.6
U3 - Beta(0.3, 0.7)						909
U3 - Beta(0.5, 0.5)						908.9
U3 - Beta(0.7, 0.3)						907.1
U3 - Beta(1, 1)						907.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	2.3	0.4	0	0	86.6
U2 - Beta(0.5, 0.5)	12	3.2	0.3	0	0	84.5
U2 - Beta(0.7, 0.3)	12.4	2.7	0.5	0	0	84.4
U2 - Beta(1, 1)	12.1	2.4	0.3	0	0	85.2
U3 - Beta(0.3, 0.7)	10.3	2.6	0.3	0	0	86.8
U3 - Beta(0.5, 0.5)	11.8	2.7	0.3	0	0	85.2
U3 - Beta(0.7, 0.3)	11.6	2.9	0.4	0	0	85.1
U3 - Beta(1, 1)	12.3	2.7	0.3	0	0	84.7
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	4.1	0.9	0.3	0	-
U2 - Beta(0.5, 0.5)	11.6	4.3	1	0.4	0	-
U2 - Beta(0.7, 0.3)	11.7	4.3	1	0.4	0.1	-
U2 - Beta(1, 1)	11.6	4.3	1	0.4	0	-
U3 - Beta(0.3, 0.7)	11.8	4	0.8	0.3	0	-
U3 - Beta(0.5, 0.5)	11.7	4.2	1	0.4	0	-
U3 - Beta(0.7, 0.3)	11.7	4.3	1	0.4	0	-
U3 - Beta(1, 1)	11.7	4.3	1	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	2.3	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3.2	2.4	0.3	0.2	0	-
U2 - Beta(0.7, 0.3)	3.3	2.4	0.3	0.2	0	-
U2 - Beta(1, 1)	3.2	2.4	0.3	0.2	0	-
U3 - Beta(0.3, 0.7)	3.4	2.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	3.3	2.4	0.3	0.2	0	-
U3 - Beta(0.7, 0.3)	3.3	2.4	0.3	0.2	0	-
U3 - Beta(1, 1)	3.3	2.4	0.3	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						594.7
U2 - Beta(0.5, 0.5)						607.2
U2 - Beta(0.7, 0.3)						609.5
U2 - Beta(1, 1)						606.7
U3 - Beta(0.3, 0.7)						592
U3 - Beta(0.5, 0.5)						606.2
U3 - Beta(0.7, 0.3)						608.3
U3 - Beta(1, 1)						607.1
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	38.6	39	18.4	0.7	0	3.3
U2 - Beta(0.5, 0.5)	32.9	25.9	32.6	5.9	0.2	2.5
U2 - Beta(0.7, 0.3)	33.8	28	28.5	6.2	0.8	2.7
U2 - Beta(1, 1)	31.8	27.1	33.3	4.8	0.3	2.7
U3 - Beta(0.3, 0.7)	46.5	37.2	12.5	0.3	0	3.5
U3 - Beta(0.5, 0.5)	35.3	27.5	29.9	4.8	0.1	2.4
U3 - Beta(0.7, 0.3)	35.5	28	27.7	6.3	0.2	2.3
U3 - Beta(1, 1)	36.1	28.4	29.2	4	0.1	2.2
Mean administrations						
U2 - Beta(0.3, 0.7)	15.1	10.4	3.9	0.2	0	-
U2 - Beta(0.5, 0.5)	11.5	8.9	7.5	1.7	0.1	-
U2 - Beta(0.7, 0.3)	11.4	8.3	6.6	2.8	0.5	-
U2 - Beta(1, 1)	11.2	9	7.7	1.6	0.1	-
U3 - Beta(0.3, 0.7)	16.3	10.3	2.8	0.1	0	-
U3 - Beta(0.5, 0.5)	12.1	9	7.1	1.4	0	-
U3 - Beta(0.7, 0.3)	11.6	8.7	6.7	2.5	0.2	-
U3 - Beta(1, 1)	12.1	9.3	6.9	1.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	4.5	1.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3	3.9	2.2	0.6	0	-
U2 - Beta(0.7, 0.3)	3.1	3.7	1.8	0.8	0.2	-
U2 - Beta(1, 1)	2.9	4	2.3	0.5	0	-
U3 - Beta(0.3, 0.7)	5.1	4.5	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	3.3	4	2.1	0.5	0	-
U3 - Beta(0.7, 0.3)	3.1	3.8	2	0.8	0.1	-
U3 - Beta(1, 1)	3.3	4.1	2	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1053.7
U2 - Beta(0.5, 0.5)						1093.5
U2 - Beta(0.7, 0.3)						1102.5
U2 - Beta(1, 1)						1095.6
U3 - Beta(0.3, 0.7)						1044.5
U3 - Beta(0.5, 0.5)						1090.5
U3 - Beta(0.7, 0.3)						1096.7
U3 - Beta(1, 1)						1090.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11	68.4	8.3	0.2	0	12.1
U2 - Beta(0.5, 0.5)	12.1	53.7	18.8	4	0.2	11.2
U2 - Beta(0.7, 0.3)	11.8	53.8	18.8	4.3	0.2	11.1
U2 - Beta(1, 1)	11.6	53	20.6	3.8	0.1	10.9
U3 - Beta(0.3, 0.7)	13.5	67.8	6.3	0.1	0	12.3
U3 - Beta(0.5, 0.5)	13.2	56.7	16.3	3	0	10.8
U3 - Beta(0.7, 0.3)	13.8	53.5	17.7	4.3	0.1	10.6
U3 - Beta(1, 1)	12.4	57.5	16.1	3.1	0	10.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	16.7	2.1	0.1	0	-
U2 - Beta(0.5, 0.5)	8.5	12.9	5.3	1.7	0.1	-
U2 - Beta(0.7, 0.3)	8.3	11.2	5.8	2.8	0.5	-
U2 - Beta(1, 1)	8.3	13	5.4	1.7	0.1	-
U3 - Beta(0.3, 0.7)	9.8	16.7	1.7	0.1	0	-
U3 - Beta(0.5, 0.5)	8.7	13.6	4.8	1.5	0.1	-
U3 - Beta(0.7, 0.3)	8.5	11.9	5.4	2.5	0.3	-
U3 - Beta(1, 1)	8.6	13.7	4.7	1.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	6.6	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.9	5.3	1.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.7	1.7	1	0.2	-
U2 - Beta(1, 1)	1.8	5.4	1.6	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.5	6.5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	2	5.6	1.4	0.6	0	-
U3 - Beta(0.7, 0.3)	1.9	5	1.6	0.8	0.1	-
U3 - Beta(1, 1)	2	5.6	1.4	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1025.8
U2 - Beta(0.5, 0.5)						1043.4
U2 - Beta(0.7, 0.3)						1051.3
U2 - Beta(1, 1)						1044.1
U3 - Beta(0.3, 0.7)						1022.6
U3 - Beta(0.5, 0.5)						1042
U3 - Beta(0.7, 0.3)						1046.1
U3 - Beta(1, 1)						1042.2
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.2	9.5	43.5	40	0.7	2.1
U2 - Beta(0.5, 0.5)	6.2	9.5	25.3	51.4	5.5	2.1
U2 - Beta(0.7, 0.3)	6.1	10.9	26.7	45.4	9	1.9
U2 - Beta(1, 1)	6.1	9.4	24.6	52.6	5.3	2
U3 - Beta(0.3, 0.7)	7.4	12.7	40.5	36.7	0.5	2.2
U3 - Beta(0.5, 0.5)	8.2	10.8	24.6	50.3	4.1	2
U3 - Beta(0.7, 0.3)	9.2	11.4	25.2	44.7	7.5	2
U3 - Beta(1, 1)	10.6	10.8	21.8	51.3	3.5	2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	6.3	11.4	7.2	0.2	-
U2 - Beta(0.5, 0.5)	4.5	5.3	8.1	10.2	1.7	-
U2 - Beta(0.7, 0.3)	4.5	5.4	7.6	9.2	3.1	-
U2 - Beta(1, 1)	4.5	5.3	8	10.3	1.7	-
U3 - Beta(0.3, 0.7)	5.1	6.8	11.2	6.5	0.1	-
U3 - Beta(0.5, 0.5)	4.7	5.5	8.1	10	1.4	-
U3 - Beta(0.7, 0.3)	4.8	5.7	7.6	9.3	2.4	-
U3 - Beta(1, 1)	4.9	5.6	7.9	10	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.5	4.1	2.4	0.1	-
U2 - Beta(0.5, 0.5)	0.6	3.3	2.4	3.3	0.6	-
U2 - Beta(0.7, 0.3)	0.6	3.4	2.2	3	0.9	-
U2 - Beta(1, 1)	0.6	3.3	2.4	3.3	0.6	-
U3 - Beta(0.3, 0.7)	1	3.7	4.1	2.2	0	-
U3 - Beta(0.5, 0.5)	0.7	3.4	2.5	3.3	0.5	-
U3 - Beta(0.7, 0.3)	0.8	3.5	2.2	3	0.7	-
U3 - Beta(1, 1)	0.8	3.5	2.3	3.3	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.3
U2 - Beta(0.5, 0.5)						1074.5
U2 - Beta(0.7, 0.3)						1082.1
U2 - Beta(1, 1)						1074.2
U3 - Beta(0.3, 0.7)						1027.4
U3 - Beta(0.5, 0.5)						1070.9
U3 - Beta(0.7, 0.3)						1074.1
U3 - Beta(1, 1)						1069.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.6	52.9	29.1	12.8	1.3	2.3
U2 - Beta(0.5, 0.5)	4.1	31.7	29.4	24.5	8.1	2.2
U2 - Beta(0.7, 0.3)	4.7	37.9	26.6	22.3	6.3	2.2
U2 - Beta(1, 1)	4	34.1	28.8	23.1	7.7	2.3
U3 - Beta(0.3, 0.7)	2.7	58.2	26.3	9.8	0.7	2.3
U3 - Beta(0.5, 0.5)	6	40.9	27.2	18.8	5	2.1
U3 - Beta(0.7, 0.3)	5.5	43.5	25.6	19.1	4.1	2.2
U3 - Beta(1, 1)	6.8	42.9	26.5	17.6	4	2.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	15.6	6.7	2.3	0.2	-
U2 - Beta(0.5, 0.5)	4.2	8.5	7.6	6.6	2.7	-
U2 - Beta(0.7, 0.3)	4.2	7.4	7.4	7.2	3.4	-
U2 - Beta(1, 1)	4.2	8.5	7.6	6.7	2.7	-
U3 - Beta(0.3, 0.7)	5	16.2	6.3	2	0.2	-
U3 - Beta(0.5, 0.5)	4.4	9.4	7.6	6.2	2.1	-
U3 - Beta(0.7, 0.3)	4.3	8.3	7.4	6.8	2.9	-
U3 - Beta(1, 1)	4.3	9.5	7.7	6.1	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	6.8	2.5	0.9	0.1	-
U2 - Beta(0.5, 0.5)	0.4	4.3	2.5	2.5	1	-
U2 - Beta(0.7, 0.3)	0.4	4	2.3	2.7	1.2	-
U2 - Beta(1, 1)	0.4	4.3	2.5	2.6	1	-
U3 - Beta(0.3, 0.7)	0.8	7.1	2.3	0.8	0.1	-
U3 - Beta(0.5, 0.5)	0.5	4.6	2.5	2.4	0.8	-
U3 - Beta(0.7, 0.3)	0.5	4.2	2.5	2.5	1.1	-
U3 - Beta(1, 1)	0.5	4.7	2.6	2.4	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1023.5
U2 - Beta(0.5, 0.5)						1039.2
U2 - Beta(0.7, 0.3)						1042.8
U2 - Beta(1, 1)						1039
U3 - Beta(0.3, 0.7)						1022.8
U3 - Beta(0.5, 0.5)						1037.1
U3 - Beta(0.7, 0.3)						1040.5
U3 - Beta(1, 1)						1035.3

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	65.5	6.3	0.4	0	0	27.8
U2 - Beta(0.5, 0.5)	63.8	8.7	1	0.3	0	26.2
U2 - Beta(0.7, 0.3)	63.3	9.9	0.7	0	0	26.1
U2 - Beta(1, 1)	63.5	9	1.2	0.3	0	26
U3 - Beta(0.3, 0.7)	66.7	5.2	0.2	0	0	27.9
U3 - Beta(0.5, 0.5)	64.3	8.1	0.5	0.3	0	26.8
U3 - Beta(0.7, 0.3)	65.4	8.1	0.5	0.1	0	25.9
U3 - Beta(1, 1)	65.2	7.8	0.5	0.2	0	26.3
Mean administrations						
U2 - Beta(0.3, 0.7)	22.7	2.7	0.2	0	0	-
U2 - Beta(0.5, 0.5)	19.8	5	1	0.2	0	-
U2 - Beta(0.7, 0.3)	19.1	5.4	1.2	0.5	0.1	-
U2 - Beta(1, 1)	19.7	5	1	0.2	0	-
U3 - Beta(0.3, 0.7)	22.9	2.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	20	4.9	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	19.5	5.1	1.1	0.4	0	-
U3 - Beta(1, 1)	20.2	4.8	0.8	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	1.2	0.1	0	0	-
U2 - Beta(0.5, 0.5)	5.2	2.1	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	5	2.2	0.3	0.2	0	-
U2 - Beta(1, 1)	5.2	2.2	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	6.5	1.2	0	0	0	-
U3 - Beta(0.5, 0.5)	5.3	2.1	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	5.1	2.1	0.4	0.1	0	-
U3 - Beta(1, 1)	5.4	2.1	0.2	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						979.5
U2 - Beta(0.5, 0.5)						1006.2
U2 - Beta(0.7, 0.3)						1008.1
U2 - Beta(1, 1)						1005.2
U3 - Beta(0.3, 0.7)						980.4
U3 - Beta(0.5, 0.5)						998.6
U3 - Beta(0.7, 0.3)						1005.4
U3 - Beta(1, 1)						1001.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.5	65	2.4	0	0	4.1
U2 - Beta(0.5, 0.5)	17.4	62.1	14.8	2.5	0.4	2.8
U2 - Beta(0.7, 0.3)	19.1	50.5	20.1	6.3	1.3	2.7
U2 - Beta(1, 1)	17.7	61.8	14.5	2.7	0.4	2.9
U3 - Beta(0.3, 0.7)	31.9	62.2	1.8	0.1	0	4
U3 - Beta(0.5, 0.5)	20	63.5	11.8	1.9	0.2	2.6
U3 - Beta(0.7, 0.3)	19.9	55.2	17.1	4.9	0.5	2.4
U3 - Beta(1, 1)	19.1	64.4	11.9	1.7	0.1	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	14.4	14.5	0.5	0	0	-
U2 - Beta(0.5, 0.5)	9.8	15.4	3.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	9.5	11.6	5.4	2.5	0.6	-
U2 - Beta(1, 1)	9.7	15.4	3.6	0.8	0.1	-
U3 - Beta(0.3, 0.7)	14.8	14.2	0.4	0	0	-
U3 - Beta(0.5, 0.5)	10.2	15.8	3	0.6	0	-
U3 - Beta(0.7, 0.3)	9.7	13.4	4.6	1.7	0.2	-
U3 - Beta(1, 1)	10.1	15.8	3	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	5.2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.2	5.6	1.1	0.2	0	-
U2 - Beta(0.7, 0.3)	2.2	4.4	1.4	0.8	0.2	-
U2 - Beta(1, 1)	2.2	5.6	1.1	0.3	0	-
U3 - Beta(0.3, 0.7)	4.4	5.1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.4	5.7	0.9	0.2	0	-
U3 - Beta(0.7, 0.3)	2.2	4.8	1.4	0.6	0.1	-
U3 - Beta(1, 1)	2.3	5.7	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1095.1
U2 - Beta(0.5, 0.5)						1130.8
U2 - Beta(0.7, 0.3)						1139.4
U2 - Beta(1, 1)						1130.4
U3 - Beta(0.3, 0.7)						1093.5
U3 - Beta(0.5, 0.5)						1129.2
U3 - Beta(0.7, 0.3)						1135.6
U3 - Beta(1, 1)						1128.3
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.8	2.8	0.4	0	0	71
U2 - Beta(0.5, 0.5)	26.6	3.4	0.3	0	0	69.7
U2 - Beta(0.7, 0.3)	27.2	2.8	0.2	0	0	69.8
U2 - Beta(1, 1)	27.1	3.5	0.1	0	0	69.3
U3 - Beta(0.3, 0.7)	26.2	2.4	0.2	0	0	71.2
U3 - Beta(0.5, 0.5)	27.2	2.8	0.3	0	0	69.7
U3 - Beta(0.7, 0.3)	27.4	2.7	0.3	0	0	69.6
U3 - Beta(1, 1)	27.5	2.6	0.1	0	0	69.8
Mean administrations						
U2 - Beta(0.3, 0.7)	14.9	3.8	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	14.3	4.3	0.8	0.2	0	-
U2 - Beta(0.7, 0.3)	14.4	4.3	0.8	0.3	0	-
U2 - Beta(1, 1)	14.3	4.3	0.8	0.3	0	-
U3 - Beta(0.3, 0.7)	15.1	3.7	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	14.5	4.2	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	14.4	4.3	0.8	0.2	0	-
U3 - Beta(1, 1)	14.5	4.3	0.7	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.3	2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	4.1	2.2	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	4.1	2.2	0.3	0.1	0	-
U2 - Beta(1, 1)	4.1	2.3	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	4.4	2	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.1	2.2	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	4.1	2.2	0.3	0.1	0	-
U3 - Beta(1, 1)	4.1	2.2	0.2	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						690.2
U2 - Beta(0.5, 0.5)						709.2
U2 - Beta(0.7, 0.3)						711
U2 - Beta(1, 1)						710.7
U3 - Beta(0.3, 0.7)						688.9
U3 - Beta(0.5, 0.5)						711.6
U3 - Beta(0.7, 0.3)						708.2
U3 - Beta(1, 1)						709.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.5	14.9	27.5	15.3	5	24.8
U2 - Beta(0.5, 0.5)	16.8	16.1	22.1	11.9	10.6	22.5
U2 - Beta(0.7, 0.3)	16.3	15.8	21.2	12.8	10.2	23.7
U2 - Beta(1, 1)	18	15.6	20.3	12.3	10.9	22.9
U3 - Beta(0.3, 0.7)	15.1	17.7	23.8	14	3.9	25.5
U3 - Beta(0.5, 0.5)	19.6	17.1	20.3	11.4	8.6	23
U3 - Beta(0.7, 0.3)	20.6	16.9	18.2	11	9.6	23.7
U3 - Beta(1, 1)	22.1	16.6	17.8	10.7	9	23.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	7	7	3.7	0.8	-
U2 - Beta(0.5, 0.5)	8.1	6.6	5.9	4.2	2.2	-
U2 - Beta(0.7, 0.3)	8.1	6.6	5.6	4.2	2.5	-
U2 - Beta(1, 1)	8	6.7	5.9	4.3	2.2	-
U3 - Beta(0.3, 0.7)	8.9	7.1	6.8	3.4	0.6	-
U3 - Beta(0.5, 0.5)	8.4	6.6	5.8	4.2	2.1	-
U3 - Beta(0.7, 0.3)	8.4	6.8	5.5	4.1	2.3	-
U3 - Beta(1, 1)	8.6	6.8	5.4	4.1	2.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.8	2.6	1.5	0.3	-
U2 - Beta(0.5, 0.5)	2	3.7	2.1	1.7	0.7	-
U2 - Beta(0.7, 0.3)	2	3.8	2	1.7	0.8	-
U2 - Beta(1, 1)	1.9	3.8	2	1.7	0.8	-
U3 - Beta(0.3, 0.7)	2.4	3.8	2.6	1.4	0.2	-
U3 - Beta(0.5, 0.5)	2.1	3.7	2	1.7	0.7	-
U3 - Beta(0.7, 0.3)	2.1	3.8	1.9	1.6	0.8	-
U3 - Beta(1, 1)	2.2	3.9	1.8	1.6	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						908.3
U2 - Beta(0.5, 0.5)						930.3
U2 - Beta(0.7, 0.3)						932.3
U2 - Beta(1, 1)						928.6
U3 - Beta(0.3, 0.7)						902.8
U3 - Beta(0.5, 0.5)						928.2
U3 - Beta(0.7, 0.3)						928.2
U3 - Beta(1, 1)						924.7

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.8	26.5	3.2	0.1	0	57.4
U2 - Beta(0.5, 0.5)	14.2	23.3	7.8	1.3	0.1	53.3
U2 - Beta(0.7, 0.3)	14.9	23.8	6.9	1.6	0.4	52.4
U2 - Beta(1, 1)	14.5	23	7.9	1.1	0.1	53.4
U3 - Beta(0.3, 0.7)	13.6	26.7	2.7	0	0	57
U3 - Beta(0.5, 0.5)	14.5	23.8	6.8	1.3	0	53.6
U3 - Beta(0.7, 0.3)	14.2	23.7	7.6	1.6	0	52.9
U3 - Beta(1, 1)	14.6	23.4	7.2	1.2	0	53.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	9.7	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	10.7	8.6	2.6	0.6	0	-
U2 - Beta(0.7, 0.3)	10.6	8	2.9	1.1	0.2	-
U2 - Beta(1, 1)	10.7	8.6	2.7	0.6	0	-
U3 - Beta(0.3, 0.7)	11.3	9.7	1	0	0	-
U3 - Beta(0.5, 0.5)	10.8	8.7	2.4	0.6	0	-
U3 - Beta(0.7, 0.3)	10.7	8.1	2.7	1	0.1	-
U3 - Beta(1, 1)	10.8	8.8	2.4	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	4.3	0.4	0	0	-
U2 - Beta(0.5, 0.5)	2.7	3.9	0.8	0.3	0	-
U2 - Beta(0.7, 0.3)	2.6	3.7	0.8	0.4	0.1	-
U2 - Beta(1, 1)	2.6	3.9	0.8	0.3	0	-
U3 - Beta(0.3, 0.7)	2.9	4.3	0.4	0	0	-
U3 - Beta(0.5, 0.5)	2.7	4	0.7	0.2	0	-
U3 - Beta(0.7, 0.3)	2.6	3.8	0.8	0.4	0	-
U3 - Beta(1, 1)	2.7	4	0.7	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						791.7
U2 - Beta(0.5, 0.5)						820.8
U2 - Beta(0.7, 0.3)						830.1
U2 - Beta(1, 1)						820.4
U3 - Beta(0.3, 0.7)						791.2
U3 - Beta(0.5, 0.5)						819
U3 - Beta(0.7, 0.3)						824.2
U3 - Beta(1, 1)						818.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	69.5	15	0.4	0	0	15.1
U2 - Beta(0.5, 0.5)	69.5	14.9	0.5	0.1	0	15
U2 - Beta(0.7, 0.3)	68.9	15.5	0.6	0.1	0	14.9
U2 - Beta(1, 1)	67.9	15.6	0.9	0.1	0	15.5
U3 - Beta(0.3, 0.7)	69.5	14.7	0.1	0	0	15.7
U3 - Beta(0.5, 0.5)	71.1	13.9	0.2	0	0	14.8
U3 - Beta(0.7, 0.3)	70.6	13.8	0.5	0.1	0	15
U3 - Beta(1, 1)	68.7	15.8	0.4	0	0	15.1
Mean administrations						
U2 - Beta(0.3, 0.7)	20.8	6	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	18.9	6.8	1.6	0.5	0	-
U2 - Beta(0.7, 0.3)	18.5	7	1.7	0.6	0	-
U2 - Beta(1, 1)	18.6	7	1.6	0.5	0	-
U3 - Beta(0.3, 0.7)	21.2	5.8	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	19.2	6.6	1.5	0.5	0	-
U3 - Beta(0.7, 0.3)	18.9	6.9	1.6	0.5	0	-
U3 - Beta(1, 1)	19	6.8	1.5	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.5	2.9	0.4	0	0	-
U2 - Beta(0.5, 0.5)	5.6	3.3	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	5.5	3.4	0.5	0.2	0	-
U2 - Beta(1, 1)	5.5	3.4	0.6	0.2	0	-
U3 - Beta(0.3, 0.7)	6.6	2.8	0.3	0	0	-
U3 - Beta(0.5, 0.5)	5.7	3.2	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	5.6	3.3	0.5	0.2	0	-
U3 - Beta(1, 1)	5.7	3.3	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						989.7
U2 - Beta(0.5, 0.5)						1002
U2 - Beta(0.7, 0.3)						1003.8
U2 - Beta(1, 1)						1001.5
U3 - Beta(0.3, 0.7)						986
U3 - Beta(0.5, 0.5)						1001.9
U3 - Beta(0.7, 0.3)						1003
U3 - Beta(1, 1)						1001.8
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10	15.5	17.7	52	2.3	2.5
U2 - Beta(0.5, 0.5)	12.8	16.2	16.2	43.3	9	2.5
U2 - Beta(0.7, 0.3)	13.4	15.6	15.9	42.6	10	2.5
U2 - Beta(1, 1)	13.4	14.6	15.6	44.6	9.2	2.6
U3 - Beta(0.3, 0.7)	14.4	19.3	14.4	46.9	2.2	2.8
U3 - Beta(0.5, 0.5)	16.2	17.5	14.3	41.8	7.8	2.4
U3 - Beta(0.7, 0.3)	16.8	18.5	13.6	39.3	9.3	2.5
U3 - Beta(1, 1)	19.3	16.1	11	42.9	8.2	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	7	7.4	9.5	0.8	-
U2 - Beta(0.5, 0.5)	5	6.3	6.5	9.1	2.9	-
U2 - Beta(0.7, 0.3)	5.1	6.2	6.4	8.4	3.7	-
U2 - Beta(1, 1)	5.1	6.2	6.4	9.1	2.9	-
U3 - Beta(0.3, 0.7)	5.6	7.5	7.1	8.8	0.6	-
U3 - Beta(0.5, 0.5)	5.4	6.4	6.3	8.9	2.7	-
U3 - Beta(0.7, 0.3)	5.6	6.5	6.1	8.2	3.3	-
U3 - Beta(1, 1)	5.7	6.5	6.1	8.9	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.9	2.7	3.4	0.3	-
U2 - Beta(0.5, 0.5)	0.8	3.8	2	3.3	0.9	-
U2 - Beta(0.7, 0.3)	0.9	3.8	1.9	3.1	1	-
U2 - Beta(1, 1)	0.9	3.8	1.9	3.3	0.9	-
U3 - Beta(0.3, 0.7)	1.2	4	2.6	3.2	0.2	-
U3 - Beta(0.5, 0.5)	1	3.9	1.9	3.2	0.8	-
U3 - Beta(0.7, 0.3)	1.2	3.9	1.8	2.9	1	-
U3 - Beta(1, 1)	1.2	3.9	1.8	3.2	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021
U2 - Beta(0.5, 0.5)						1046.7
U2 - Beta(0.7, 0.3)						1051.4
U2 - Beta(1, 1)						1043.8
U3 - Beta(0.3, 0.7)						1013.6
U3 - Beta(0.5, 0.5)						1041.5
U3 - Beta(0.7, 0.3)						1042.3
U3 - Beta(1, 1)						1040.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.2	24.8	6.6	1.1	0	54.3
U2 - Beta(0.5, 0.5)	14.2	24.7	7.9	1.7	0.1	51.4
U2 - Beta(0.7, 0.3)	14.9	25.6	7.1	1.6	0.2	50.6
U2 - Beta(1, 1)	14.8	24.4	7.5	1.6	0.1	51.6
U3 - Beta(0.3, 0.7)	13.9	25.9	5.7	0.8	0	53.7
U3 - Beta(0.5, 0.5)	14.9	25.7	6.4	1.3	0.1	51.6
U3 - Beta(0.7, 0.3)	15.4	26.2	5.8	1.4	0.1	51.1
U3 - Beta(1, 1)	15.8	26	5.2	1.5	0.1	51.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	9.4	1.9	0.3	0	-
U2 - Beta(0.5, 0.5)	10.6	8.1	3	1.2	0.2	-
U2 - Beta(0.7, 0.3)	10.5	7.9	3.1	1.4	0.3	-
U2 - Beta(1, 1)	10.6	8.2	3	1.2	0.2	-
U3 - Beta(0.3, 0.7)	11.2	9.5	1.7	0.3	0	-
U3 - Beta(0.5, 0.5)	10.7	8.3	2.9	1.1	0.1	-
U3 - Beta(0.7, 0.3)	10.6	8.1	3	1.3	0.2	-
U3 - Beta(1, 1)	10.7	8.4	2.8	1.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	4.4	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	3.9	1	0.5	0.1	-
U2 - Beta(0.7, 0.3)	2.7	3.8	0.9	0.6	0.1	-
U2 - Beta(1, 1)	2.7	3.9	1	0.5	0.1	-
U3 - Beta(0.3, 0.7)	3	4.4	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	2.8	4	0.9	0.5	0.1	-
U3 - Beta(0.7, 0.3)	2.7	3.9	0.9	0.5	0.1	-
U3 - Beta(1, 1)	2.8	4.1	0.9	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						796.2
U2 - Beta(0.5, 0.5)						817.6
U2 - Beta(0.7, 0.3)						825.1
U2 - Beta(1, 1)						819.4
U3 - Beta(0.3, 0.7)						795.3
U3 - Beta(0.5, 0.5)						818
U3 - Beta(0.7, 0.3)						822.3
U3 - Beta(1, 1)						818
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.5	21.8	23.3	29.5	3.5	8.4
U2 - Beta(0.5, 0.5)	20	21.3	20.3	23.7	5.9	8.8
U2 - Beta(0.7, 0.3)	21.3	19.3	19.6	24	6.4	9.4
U2 - Beta(1, 1)	20.1	20.1	19.7	24.8	6.4	8.9
U3 - Beta(0.3, 0.7)	20.7	24.3	15.4	27.7	2.8	9.1
U3 - Beta(0.5, 0.5)	25	24	16	21.4	4.7	8.9
U3 - Beta(0.7, 0.3)	25.7	23.6	16.3	20	5	9.4
U3 - Beta(1, 1)	25.5	23	15.6	22	4.5	9.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6.8	7.8	7.3	6.3	0.9	-
U2 - Beta(0.5, 0.5)	6.7	7.3	6.6	6.4	2.2	-
U2 - Beta(0.7, 0.3)	6.7	7.2	6.5	6.1	2.5	-
U2 - Beta(1, 1)	6.8	7.2	6.5	6.3	2.3	-
U3 - Beta(0.3, 0.7)	7.5	8	6.7	6	0.8	-
U3 - Beta(0.5, 0.5)	7.2	7.3	6.4	6.2	2.1	-
U3 - Beta(0.7, 0.3)	7.2	7.4	6.2	5.9	2.3	-
U3 - Beta(1, 1)	7.4	7.5	6	6.1	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	4.2	2.7	2.5	0.3	-
U2 - Beta(0.5, 0.5)	1.5	4.1	2.2	2.5	0.7	-
U2 - Beta(0.7, 0.3)	1.5	4.1	2.1	2.5	0.8	-
U2 - Beta(1, 1)	1.6	4.1	2.1	2.5	0.8	-
U3 - Beta(0.3, 0.7)	2	4.2	2.5	2.3	0.3	-
U3 - Beta(0.5, 0.5)	1.8	4.2	2.1	2.4	0.7	-
U3 - Beta(0.7, 0.3)	1.8	4.2	2	2.3	0.8	-
U3 - Beta(1, 1)	1.9	4.3	1.9	2.4	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						974
U2 - Beta(0.5, 0.5)						988.1
U2 - Beta(0.7, 0.3)						988.4
U2 - Beta(1, 1)						987.1
U3 - Beta(0.3, 0.7)						969.9
U3 - Beta(0.5, 0.5)						986.6
U3 - Beta(0.7, 0.3)						984.8
U3 - Beta(1, 1)						983.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	38.8	33.4	11.2	1.9	0	14.7
U2 - Beta(0.5, 0.5)	41.9	31.5	9.6	1.9	0.1	15
U2 - Beta(0.7, 0.3)	40.3	32.6	10.3	1.6	0	15.2
U2 - Beta(1, 1)	39.8	32.2	10.9	1.7	0	15.4
U3 - Beta(0.3, 0.7)	45	31.8	6.2	1.2	0	15.8
U3 - Beta(0.5, 0.5)	45.5	30.3	7.4	1.2	0.1	15.5
U3 - Beta(0.7, 0.3)	43.6	31.4	8.4	1.1	0.1	15.4
U3 - Beta(1, 1)	44.4	30.5	7.8	1.6	0.1	15.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.4	9.7	4.8	2.1	0.4	-
U2 - Beta(0.5, 0.5)	11.3	9	4.8	2.6	0.6	-
U2 - Beta(0.7, 0.3)	11.4	8.8	4.9	2.7	0.7	-
U2 - Beta(1, 1)	11.1	9	5	2.7	0.7	-
U3 - Beta(0.3, 0.7)	12.5	9.5	4.1	1.7	0.3	-
U3 - Beta(0.5, 0.5)	11.8	8.8	4.6	2.6	0.6	-
U3 - Beta(0.7, 0.3)	11.8	8.8	4.6	2.6	0.6	-
U3 - Beta(1, 1)	11.9	8.9	4.5	2.5	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	4.7	1.9	0.9	0.2	-
U2 - Beta(0.5, 0.5)	3.3	4.6	1.7	1.2	0.3	-
U2 - Beta(0.7, 0.3)	3.4	4.5	1.7	1.2	0.3	-
U2 - Beta(1, 1)	3.3	4.6	1.7	1.2	0.3	-
U3 - Beta(0.3, 0.7)	4	4.6	1.6	0.7	0.1	-
U3 - Beta(0.5, 0.5)	3.6	4.5	1.6	1.1	0.3	-
U3 - Beta(0.7, 0.3)	3.6	4.5	1.6	1.1	0.2	-
U3 - Beta(1, 1)	3.6	4.6	1.5	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						946.2
U2 - Beta(0.5, 0.5)						953
U2 - Beta(0.7, 0.3)						953.3
U2 - Beta(1, 1)						954.3
U3 - Beta(0.3, 0.7)						939.4
U3 - Beta(0.5, 0.5)						952.2
U3 - Beta(0.7, 0.3)						952
U3 - Beta(1, 1)						951.6

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.7	27.9	23.7	17.7	9	0
U2 - Beta(0.5, 0.5)	31	30.6	16.4	13.8	8.2	0
U2 - Beta(0.7, 0.3)	31.4	34.3	15.9	12.1	6.3	0
U2 - Beta(1, 1)	33.8	30.8	16.4	12.6	6.4	0
U3 - Beta(0.3, 0.7)	28.4	32.3	20.8	13.7	4.8	0
U3 - Beta(0.5, 0.5)	36.5	33.7	14.7	10.9	4.2	0
U3 - Beta(0.7, 0.3)	35	36.6	13.8	10.3	4.3	0
U3 - Beta(1, 1)	39.9	33.2	13.2	9.3	4.4	0
Mean administrations						
U2 - Beta(0.3, 0.7)	10	9.3	5.6	3.6	1.6	-
U2 - Beta(0.5, 0.5)	4.4	5.6	6.2	8.1	5.7	-
U2 - Beta(0.7, 0.3)	4.2	5.2	6	8.6	6	-
U2 - Beta(1, 1)	4.4	5.7	6.4	8	5.5	-
U3 - Beta(0.3, 0.7)	10.7	9.8	5.2	3.1	1.2	-
U3 - Beta(0.5, 0.5)	4.9	6.2	6.3	7.7	5	-
U3 - Beta(0.7, 0.3)	4.8	5.9	6.1	8.1	5.2	-
U3 - Beta(1, 1)	5	6.4	6.4	7.5	4.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	4.4	2.4	1.5	0.7	-
U2 - Beta(0.5, 0.5)	0.6	3.1	2.4	3.5	2.2	-
U2 - Beta(0.7, 0.3)	0.5	3	2.2	3.7	2.2	-
U2 - Beta(1, 1)	0.6	3.1	2.5	3.4	2.1	-
U3 - Beta(0.3, 0.7)	3.4	4.6	2.3	1.3	0.5	-
U3 - Beta(0.5, 0.5)	0.8	3.3	2.4	3.3	1.9	-
U3 - Beta(0.7, 0.3)	0.7	3.3	2.3	3.5	1.9	-
U3 - Beta(1, 1)	0.8	3.4	2.5	3.2	1.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						986.8
U2 - Beta(0.5, 0.5)						1006.7
U2 - Beta(0.7, 0.3)						1010.1
U2 - Beta(1, 1)						1007.1
U3 - Beta(0.3, 0.7)						983.8
U3 - Beta(0.5, 0.5)						1004.7
U3 - Beta(0.7, 0.3)						1006.5
U3 - Beta(1, 1)						1004.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	55.9	42.9	0.7	0	0	0.5
U2 - Beta(0.5, 0.5)	22.5	70.1	7	0.1	0	0.3
U2 - Beta(0.7, 0.3)	23.5	63.8	11.7	0.5	0	0.5
U2 - Beta(1, 1)	20.7	71.7	7.2	0.1	0	0.3
U3 - Beta(0.3, 0.7)	58.6	40.5	0.4	0	0	0.5
U3 - Beta(0.5, 0.5)	24.2	70.3	5.1	0.1	0	0.3
U3 - Beta(0.7, 0.3)	24.9	63.7	10.6	0.4	0	0.4
U3 - Beta(1, 1)	23.6	71.1	4.9	0.1	0	0.3
Mean administrations						
U2 - Beta(0.3, 0.7)	20.1	9.6	0.2	0	0	-
U2 - Beta(0.5, 0.5)	10.9	16.6	2.3	0.2	0	-
U2 - Beta(0.7, 0.3)	10.1	14.1	4.5	1.1	0.1	-
U2 - Beta(1, 1)	10.7	16.7	2.3	0.2	0	-
U3 - Beta(0.3, 0.7)	20.4	9.4	0.1	0	0	-
U3 - Beta(0.5, 0.5)	11.4	16.7	1.7	0.1	0	-
U3 - Beta(0.7, 0.3)	10.6	15.2	3.6	0.6	0	-
U3 - Beta(1, 1)	11.4	16.7	1.7	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.3	3.4	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.5	5.6	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	2.3	4.7	1.2	0.4	0	-
U2 - Beta(1, 1)	2.4	5.6	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	6.5	3.3	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.7	5.6	0.5	0.1	0	-
U3 - Beta(0.7, 0.3)	2.4	5.1	1	0.2	0	-
U3 - Beta(1, 1)	2.7	5.6	0.5	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.4
U2 - Beta(0.5, 0.5)						1165.1
U2 - Beta(0.7, 0.3)						1179
U2 - Beta(1, 1)						1166.4
U3 - Beta(0.3, 0.7)						1108
U3 - Beta(0.5, 0.5)						1162.1
U3 - Beta(0.7, 0.3)						1168.7
U3 - Beta(1, 1)						1161.6
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4	9	15.5	33.2	38.1	0.2
U2 - Beta(0.5, 0.5)	9.3	12.1	13.4	20.1	44.9	0.2
U2 - Beta(0.7, 0.3)	9.3	14.1	13.4	19.2	43.9	0.1
U2 - Beta(1, 1)	11.1	11.2	12.6	18.9	46	0.2
U3 - Beta(0.3, 0.7)	7.8	11.6	16.9	34.9	28.7	0.1
U3 - Beta(0.5, 0.5)	14.2	13.6	11.8	17.8	42.4	0.2
U3 - Beta(0.7, 0.3)	15	16.1	12.2	17	39.5	0.2
U3 - Beta(1, 1)	17.8	15.3	10.7	15.7	40.3	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	5.4	7	8.6	5	-
U2 - Beta(0.5, 0.5)	3.5	4.2	5.4	7.9	8.9	-
U2 - Beta(0.7, 0.3)	3.5	4.2	5.5	7.6	9.2	-
U2 - Beta(1, 1)	3.5	4.3	5.5	7.8	9	-
U3 - Beta(0.3, 0.7)	4	5.8	7.3	8.7	4.2	-
U3 - Beta(0.5, 0.5)	3.5	4.4	5.6	7.9	8.5	-
U3 - Beta(0.7, 0.3)	3.7	4.5	5.6	7.7	8.6	-
U3 - Beta(1, 1)	3.8	4.6	5.5	7.8	8.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3.2	2.9	3.4	1.7	-
U2 - Beta(0.5, 0.5)	0.2	2.9	1.9	3.2	2.7	-
U2 - Beta(0.7, 0.3)	0.2	2.9	1.8	3.1	2.7	-
U2 - Beta(1, 1)	0.2	2.9	1.9	3.1	2.7	-
U3 - Beta(0.3, 0.7)	0.5	3.4	3	3.5	1.4	-
U3 - Beta(0.5, 0.5)	0.2	3	1.9	3.2	2.6	-
U3 - Beta(0.7, 0.3)	0.3	3	1.9	3.1	2.6	-
U3 - Beta(1, 1)	0.4	3.1	1.9	3.1	2.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.7
U2 - Beta(0.5, 0.5)						1052
U2 - Beta(0.7, 0.3)						1053.2
U2 - Beta(1, 1)						1050.3
U3 - Beta(0.3, 0.7)						999
U3 - Beta(0.5, 0.5)						1046.9
U3 - Beta(0.7, 0.3)						1048.4
U3 - Beta(1, 1)						1044.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.7	57.7	9.1	0.1	0	17.4
U2 - Beta(0.5, 0.5)	17.2	55.1	9.8	0.2	0	17.7
U2 - Beta(0.7, 0.3)	16.5	56.6	8.9	0.6	0	17.4
U2 - Beta(1, 1)	16	55.9	10.2	0.4	0	17.5
U3 - Beta(0.3, 0.7)	18	58.4	6.8	0.1	0	16.7
U3 - Beta(0.5, 0.5)	18.6	55.2	8.6	0.2	0	17.4
U3 - Beta(0.7, 0.3)	18.9	55.8	8.1	0.3	0	16.9
U3 - Beta(1, 1)	17.3	56.9	8.7	0.3	0	16.8
Mean administrations						
U2 - Beta(0.3, 0.7)	9.7	15.4	2.8	0.2	0	-
U2 - Beta(0.5, 0.5)	9.6	13	4.6	1.1	0.1	-
U2 - Beta(0.7, 0.3)	9.4	12.5	4.7	1.6	0.1	-
U2 - Beta(1, 1)	9.3	13.1	4.7	1	0.1	-
U3 - Beta(0.3, 0.7)	10.2	15.3	2.4	0.2	0	-
U3 - Beta(0.5, 0.5)	9.7	13.3	4.3	0.9	0.1	-
U3 - Beta(0.7, 0.3)	9.6	12.7	4.5	1.4	0.1	-
U3 - Beta(1, 1)	9.6	13.3	4.3	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	6.5	1	0.1	0	-
U2 - Beta(0.5, 0.5)	2.3	5.7	1.5	0.5	0	-
U2 - Beta(0.7, 0.3)	2.2	5.6	1.3	0.6	0	-
U2 - Beta(1, 1)	2.2	5.8	1.5	0.4	0	-
U3 - Beta(0.3, 0.7)	2.7	6.5	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	2.4	5.9	1.3	0.4	0	-
U3 - Beta(0.7, 0.3)	2.4	5.6	1.4	0.6	0	-
U3 - Beta(1, 1)	2.4	5.8	1.3	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						988.9
U2 - Beta(0.5, 0.5)						1003.2
U2 - Beta(0.7, 0.3)						1010
U2 - Beta(1, 1)						1004.5
U3 - Beta(0.3, 0.7)						982.9
U3 - Beta(0.5, 0.5)						1000.6
U3 - Beta(0.7, 0.3)						1004.7
U3 - Beta(1, 1)						999.9
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.4	57.4	28	3.9	0.1	3.2
U2 - Beta(0.5, 0.5)	9	57	25.6	5.6	0	2.8
U2 - Beta(0.7, 0.3)	9.3	53.7	28	5.8	0.1	3.1
U2 - Beta(1, 1)	9.2	54.6	27.4	6.1	0	2.7
U3 - Beta(0.3, 0.7)	8.4	61.2	24.3	3.1	0	3
U3 - Beta(0.5, 0.5)	11	57.9	22.8	5.6	0	2.7
U3 - Beta(0.7, 0.3)	11.5	57.6	22.2	5.8	0	2.9
U3 - Beta(1, 1)	12	59.8	19.9	5.5	0.1	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	15.1	7.1	1.7	0.1	-
U2 - Beta(0.5, 0.5)	5.3	11.8	8.2	3.6	0.7	-
U2 - Beta(0.7, 0.3)	5.4	11.6	7.9	3.9	0.9	-
U2 - Beta(1, 1)	5.4	11.9	8.1	3.6	0.7	-
U3 - Beta(0.3, 0.7)	5.8	15.9	6.4	1.4	0.1	-
U3 - Beta(0.5, 0.5)	5.6	12.4	7.7	3.3	0.6	-
U3 - Beta(0.7, 0.3)	5.7	12	7.5	3.6	0.8	-
U3 - Beta(1, 1)	5.8	12.5	7.4	3.3	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	6.6	2.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	0.9	5.5	2.7	1.4	0.3	-
U2 - Beta(0.7, 0.3)	0.9	5.5	2.5	1.5	0.3	-
U2 - Beta(1, 1)	0.9	5.5	2.6	1.4	0.3	-
U3 - Beta(0.3, 0.7)	1.2	6.9	2.4	0.6	0	-
U3 - Beta(0.5, 0.5)	1	5.7	2.5	1.3	0.2	-
U3 - Beta(0.7, 0.3)	1	5.6	2.4	1.4	0.3	-
U3 - Beta(1, 1)	1.1	5.8	2.4	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1018.5
U2 - Beta(0.5, 0.5)						1037.3
U2 - Beta(0.7, 0.3)						1039.3
U2 - Beta(1, 1)						1036.5
U3 - Beta(0.3, 0.7)						1015.8
U3 - Beta(0.5, 0.5)						1032.9
U3 - Beta(0.7, 0.3)						1036.1
U3 - Beta(1, 1)						1032.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.9	0	0	0	0	59.1
U2 - Beta(0.5, 0.5)	37	0.1	0	0	0	62.9
U2 - Beta(0.7, 0.3)	33.8	0.1	0	0	0	66.1
U2 - Beta(1, 1)	37.7	0	0	0	0	62.3
U3 - Beta(0.3, 0.7)	40.9	0	0	0	0	59.1
U3 - Beta(0.5, 0.5)	37.7	0	0	0	0	62.3
U3 - Beta(0.7, 0.3)	36.2	0.1	0	0	0	63.7
U3 - Beta(1, 1)	37.5	0	0	0	0	62.5
Mean administrations						
U2 - Beta(0.3, 0.7)	20.2	0.3	0	0	0	-
U2 - Beta(0.5, 0.5)	18.2	1.7	0	0	0	-
U2 - Beta(0.7, 0.3)	16.7	2.9	0	0	0	-
U2 - Beta(1, 1)	18.3	1.7	0	0	0	-
U3 - Beta(0.3, 0.7)	20.2	0.3	0	0	0	-
U3 - Beta(0.5, 0.5)	18.3	1.6	0	0	0	-
U3 - Beta(0.7, 0.3)	17.6	2.2	0	0	0	-
U3 - Beta(1, 1)	18.3	1.6	0	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.3	0.2	0	0	0	-
U2 - Beta(0.5, 0.5)	3.5	0.8	0	0	0	-
U2 - Beta(0.7, 0.3)	3.1	1	0	0	0	-
U2 - Beta(1, 1)	3.5	0.8	0	0	0	-
U3 - Beta(0.3, 0.7)	4.3	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.5	0.8	0	0	0	-
U3 - Beta(0.7, 0.3)	3.3	1	0	0	0	-
U3 - Beta(1, 1)	3.5	0.8	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						878.2
U2 - Beta(0.5, 0.5)						858.9
U2 - Beta(0.7, 0.3)						846.9
U2 - Beta(1, 1)						860.8
U3 - Beta(0.3, 0.7)						879.7
U3 - Beta(0.5, 0.5)						861.1
U3 - Beta(0.7, 0.3)						858.9
U3 - Beta(1, 1)						861.6

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.1	62.4	6.5	0.3	0	12.7
U2 - Beta(0.5, 0.5)	19.8	58.9	6.9	0.2	0	14.2
U2 - Beta(0.7, 0.3)	18.4	59.1	7.1	0.5	0	14.9
U2 - Beta(1, 1)	19.2	58.7	8	0.3	0	13.8
U3 - Beta(0.3, 0.7)	19.8	61.3	5.8	0	0	13.1
U3 - Beta(0.5, 0.5)	21.5	57.4	6.8	0.3	0	14
U3 - Beta(0.7, 0.3)	20.1	59	6.8	0.4	0	13.7
U3 - Beta(1, 1)	21.2	56.9	7.1	0.5	0	14.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	15.8	3	0.4	0	-
U2 - Beta(0.5, 0.5)	9.7	13.5	4.3	1.1	0	-
U2 - Beta(0.7, 0.3)	9.8	13	4.5	1.5	0	-
U2 - Beta(1, 1)	9.7	13.5	4.3	1.2	0	-
U3 - Beta(0.3, 0.7)	10.1	15.8	2.6	0.2	0	-
U3 - Beta(0.5, 0.5)	10	13.5	4.1	1	0	-
U3 - Beta(0.7, 0.3)	9.9	13.2	4.4	1.3	0	-
U3 - Beta(1, 1)	10	13.4	4.2	1.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	6.7	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	2.5	6.1	1.3	0.5	0	-
U2 - Beta(0.7, 0.3)	2.5	6	1.2	0.6	0	-
U2 - Beta(1, 1)	2.5	6.1	1.3	0.5	0	-
U3 - Beta(0.3, 0.7)	2.8	6.7	1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.6	6.1	1.2	0.4	0	-
U3 - Beta(0.7, 0.3)	2.6	6	1.2	0.5	0	-
U3 - Beta(1, 1)	2.6	6.1	1.2	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						999.6
U2 - Beta(0.5, 0.5)						1006.4
U2 - Beta(0.7, 0.3)						1010.9
U2 - Beta(1, 1)						1006.1
U3 - Beta(0.3, 0.7)						992.9
U3 - Beta(0.5, 0.5)						1002.1
U3 - Beta(0.7, 0.3)						1007.6
U3 - Beta(1, 1)						1003.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.5	0.1	0	0	0	94.4
U2 - Beta(0.5, 0.5)	5.3	0.1	0	0	0	94.6
U2 - Beta(0.7, 0.3)	5.2	0	0	0	0	94.8
U2 - Beta(1, 1)	5.5	0.2	0	0	0	94.3
U3 - Beta(0.3, 0.7)	5.7	0	0	0	0	94.3
U3 - Beta(0.5, 0.5)	5.6	0.1	0	0	0	94.3
U3 - Beta(0.7, 0.3)	5.4	0	0	0	0	94.6
U3 - Beta(1, 1)	5.4	0.1	0	0	0	94.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	2.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	10.7	3.2	0.2	0	0	-
U2 - Beta(0.7, 0.3)	10.7	3.2	0.2	0	0	-
U2 - Beta(1, 1)	10.8	3.2	0.2	0	0	-
U3 - Beta(0.3, 0.7)	11.1	2.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	10.8	3.2	0.2	0	0	-
U3 - Beta(0.7, 0.3)	10.8	3.2	0.2	0	0	-
U3 - Beta(1, 1)	10.8	3.2	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	1.8	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.8	1.9	0.1	0	0	-
U2 - Beta(0.7, 0.3)	2.8	1.9	0.1	0	0	-
U2 - Beta(1, 1)	2.9	1.9	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3	1.7	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	1.9	0.1	0	0	-
U3 - Beta(0.7, 0.3)	2.9	1.9	0.1	0	0	-
U3 - Beta(1, 1)	2.9	1.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						505.3
U2 - Beta(0.5, 0.5)						501.7
U2 - Beta(0.7, 0.3)						503.5
U2 - Beta(1, 1)						503.7
U3 - Beta(0.3, 0.7)						504.1
U3 - Beta(0.5, 0.5)						505
U3 - Beta(0.7, 0.3)						504.3
U3 - Beta(1, 1)						504.1
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.4	1.4	15.5	74.7	8	0
U2 - Beta(0.5, 0.5)	0.2	0.5	1.6	64.6	33.1	0
U2 - Beta(0.7, 0.3)	0.7	2.1	3.9	52.2	41.1	0
U2 - Beta(1, 1)	0.3	0.7	1.5	64.3	33.2	0
U3 - Beta(0.3, 0.7)	0.6	1.3	18.4	74.7	5	0
U3 - Beta(0.5, 0.5)	0.7	1	1.8	68.3	28.2	0
U3 - Beta(0.7, 0.3)	1.8	3.2	4.1	54	36.9	0
U3 - Beta(1, 1)	0.9	1.3	2	69.5	26.3	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4.6	7.9	12.8	1.1	-
U2 - Beta(0.5, 0.5)	3.1	3.2	3.8	14	5.9	-
U2 - Beta(0.7, 0.3)	3.1	3.1	3.6	11.5	8.8	-
U2 - Beta(1, 1)	3.1	3.2	3.8	14	5.9	-
U3 - Beta(0.3, 0.7)	3.7	4.6	8.3	12.7	0.7	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	14.5	5.3	-
U3 - Beta(0.7, 0.3)	3.1	3.2	3.8	11.8	8.1	-
U3 - Beta(1, 1)	3.1	3.2	3.9	14.7	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.9	3.2	4.4	0.4	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.8	1.8	-
U2 - Beta(0.7, 0.3)	0	2.4	1.1	4	2.6	-
U2 - Beta(1, 1)	0	2.5	1.2	4.8	1.8	-
U3 - Beta(0.3, 0.7)	0.4	2.9	3.4	4.4	0.3	-
U3 - Beta(0.5, 0.5)	0	2.5	1.2	5	1.6	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.2	4.1	2.4	-
U3 - Beta(1, 1)	0	2.5	1.2	5.1	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.2
U2 - Beta(0.5, 0.5)						1087.8
U2 - Beta(0.7, 0.3)						1096.3
U2 - Beta(1, 1)						1087.2
U3 - Beta(0.3, 0.7)						1029.1
U3 - Beta(0.5, 0.5)						1085.7
U3 - Beta(0.7, 0.3)						1091
U3 - Beta(1, 1)						1085.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	30.6	48.4	19.3	0.3	0	1.4
U2 - Beta(0.5, 0.5)	28.8	40.6	28.4	1.2	0	1
U2 - Beta(0.7, 0.3)	30.8	42.1	24.4	1.5	0	1.2
U2 - Beta(1, 1)	29.6	38.4	29.6	1.2	0	1.2
U3 - Beta(0.3, 0.7)	36.2	48	14.2	0.1	0	1.5
U3 - Beta(0.5, 0.5)	32.9	40.3	25.2	0.6	0	1
U3 - Beta(0.7, 0.3)	32.9	42.4	22	1.4	0	1.3
U3 - Beta(1, 1)	32.5	40.7	24.6	1.1	0	1.1
Mean administrations						
U2 - Beta(0.3, 0.7)	12.8	12.4	4.4	0.3	0	-
U2 - Beta(0.5, 0.5)	9.5	11.2	7.7	1.4	0.1	-
U2 - Beta(0.7, 0.3)	10	10.8	6.8	2.1	0.1	-
U2 - Beta(1, 1)	9.5	11.1	7.8	1.4	0.1	-
U3 - Beta(0.3, 0.7)	14	12.5	3.3	0.2	0	-
U3 - Beta(0.5, 0.5)	10.2	11.4	7.2	1.1	0	-
U3 - Beta(0.7, 0.3)	10.2	11.1	6.7	1.9	0.1	-
U3 - Beta(1, 1)	10.3	11.3	7	1.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4	5.2	1.5	0.1	0	-
U2 - Beta(0.5, 0.5)	2.4	4.7	2.4	0.5	0	-
U2 - Beta(0.7, 0.3)	2.7	4.7	1.8	0.8	0	-
U2 - Beta(1, 1)	2.4	4.7	2.4	0.5	0	-
U3 - Beta(0.3, 0.7)	4.4	5.3	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.7	4.9	2.2	0.5	0	-
U3 - Beta(0.7, 0.3)	2.7	4.7	1.9	0.7	0	-
U3 - Beta(1, 1)	2.8	4.8	2.1	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1049
U2 - Beta(0.5, 0.5)						1090.1
U2 - Beta(0.7, 0.3)						1095.1
U2 - Beta(1, 1)						1088.6
U3 - Beta(0.3, 0.7)						1042.1
U3 - Beta(0.5, 0.5)						1084
U3 - Beta(0.7, 0.3)						1087.6
U3 - Beta(1, 1)						1082.3
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.4	39.4	21.9	0.5	0	15.8
U2 - Beta(0.5, 0.5)	26.5	35.4	20.9	1.6	0	15.6
U2 - Beta(0.7, 0.3)	26.8	37.3	18	2	0	15.9
U2 - Beta(1, 1)	24.7	36.7	21.8	1.8	0	15
U3 - Beta(0.3, 0.7)	27.2	39.7	17.2	0.5	0	15.4
U3 - Beta(0.5, 0.5)	29	34.8	19.2	1.6	0	15.4
U3 - Beta(0.7, 0.3)	28.7	37.4	15.7	2.1	0	16.1
U3 - Beta(1, 1)	26.5	37	18.7	2	0	15.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.6	11.4	5.4	0.5	0	-
U2 - Beta(0.5, 0.5)	10.4	9.8	6.2	1.6	0.1	-
U2 - Beta(0.7, 0.3)	10.6	9.8	5.6	2	0.2	-
U2 - Beta(1, 1)	10.2	10.1	6.2	1.7	0.1	-
U3 - Beta(0.3, 0.7)	11.6	11.4	4.6	0.3	0	-
U3 - Beta(0.5, 0.5)	10.7	9.8	6	1.5	0.1	-
U3 - Beta(0.7, 0.3)	10.8	9.9	5.5	1.9	0.1	-
U3 - Beta(1, 1)	10.6	10	5.9	1.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	5.1	1.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.8	4.7	1.9	0.7	0	-
U2 - Beta(0.7, 0.3)	2.9	4.7	1.6	0.8	0.1	-
U2 - Beta(1, 1)	2.7	4.8	1.9	0.7	0	-
U3 - Beta(0.3, 0.7)	3.4	5.2	1.7	0.1	0	-
U3 - Beta(0.5, 0.5)	2.9	4.7	1.8	0.6	0	-
U3 - Beta(0.7, 0.3)	2.9	4.7	1.6	0.7	0	-
U3 - Beta(1, 1)	2.9	4.7	1.8	0.7	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						967.8
U2 - Beta(0.5, 0.5)						991.3
U2 - Beta(0.7, 0.3)						999.7
U2 - Beta(1, 1)						994.5
U3 - Beta(0.3, 0.7)						964.6
U3 - Beta(0.5, 0.5)						989.8
U3 - Beta(0.7, 0.3)						994.7
U3 - Beta(1, 1)						988.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.2	37	40.3	1.7	0	6.8
U2 - Beta(0.5, 0.5)	17.2	33.7	39	3.7	0	6.4
U2 - Beta(0.7, 0.3)	18.6	34.7	35.9	3.9	0	6.9
U2 - Beta(1, 1)	17.9	32.4	39.8	3.6	0	6.3
U3 - Beta(0.3, 0.7)	19.3	39.7	32.6	1.4	0	7
U3 - Beta(0.5, 0.5)	19.3	34.4	36.7	2.9	0	6.7
U3 - Beta(0.7, 0.3)	20.4	35.8	32.9	4.3	0	6.6
U3 - Beta(1, 1)	20.9	33	36.1	3.4	0	6.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	11.9	8.3	0.7	0	-
U2 - Beta(0.5, 0.5)	7.7	9.7	9.4	2.3	0.2	-
U2 - Beta(0.7, 0.3)	7.9	9.6	8.3	3	0.4	-
U2 - Beta(1, 1)	7.6	9.7	9.4	2.3	0.2	-
U3 - Beta(0.3, 0.7)	9.3	12.3	7.1	0.4	0	-
U3 - Beta(0.5, 0.5)	8	9.9	9.1	2.2	0.2	-
U3 - Beta(0.7, 0.3)	8.1	9.8	8.2	2.9	0.2	-
U3 - Beta(1, 1)	8.1	9.9	9	2.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	5.3	2.8	0.3	0	-
U2 - Beta(0.5, 0.5)	1.8	4.5	2.8	0.9	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.6	2.4	1	0.1	-
U2 - Beta(1, 1)	1.7	4.6	2.8	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.6	5.5	2.4	0.2	0	-
U3 - Beta(0.5, 0.5)	1.9	4.6	2.7	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2	4.6	2.4	1	0.1	-
U3 - Beta(1, 1)	2	4.7	2.7	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022.4
U2 - Beta(0.5, 0.5)						1051.4
U2 - Beta(0.7, 0.3)						1054.5
U2 - Beta(1, 1)						1049.9
U3 - Beta(0.3, 0.7)						1011.7
U3 - Beta(0.5, 0.5)						1046.3
U3 - Beta(0.7, 0.3)						1049.8
U3 - Beta(1, 1)						1043.6
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.8	28.2	1	0	0	56
U2 - Beta(0.5, 0.5)	18.4	23.4	1.5	0	0	56.7
U2 - Beta(0.7, 0.3)	17.5	25	1.7	0	0	55.8
U2 - Beta(1, 1)	17	24.8	1.5	0	0	56.7
U3 - Beta(0.3, 0.7)	14.9	27.6	1.1	0	0	56.4
U3 - Beta(0.5, 0.5)	17.7	24.3	1.5	0	0	56.5
U3 - Beta(0.7, 0.3)	17.6	24.8	1.4	0	0	56.2
U3 - Beta(1, 1)	18	25.2	1.2	0	0	55.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.6	10.1	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	11.7	8.8	2.1	0.4	0	-
U2 - Beta(0.7, 0.3)	11.5	8.6	2.2	0.6	0	-
U2 - Beta(1, 1)	11.5	9	2.1	0.4	0	-
U3 - Beta(0.3, 0.7)	11.6	10.1	1	0.1	0	-
U3 - Beta(0.5, 0.5)	11.6	8.9	2	0.4	0	-
U3 - Beta(0.7, 0.3)	11.6	8.7	2.1	0.5	0	-
U3 - Beta(1, 1)	11.7	9	2	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	4.7	0.4	0	0	-
U2 - Beta(0.5, 0.5)	3.1	4.1	0.7	0.2	0	-
U2 - Beta(0.7, 0.3)	3.1	4.1	0.7	0.3	0	-
U2 - Beta(1, 1)	3	4.2	0.7	0.2	0	-
U3 - Beta(0.3, 0.7)	3.2	4.6	0.4	0	0	-
U3 - Beta(0.5, 0.5)	3.1	4.2	0.6	0.2	0	-
U3 - Beta(0.7, 0.3)	3.1	4.1	0.7	0.2	0	-
U3 - Beta(1, 1)	3.1	4.2	0.6	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						798.9
U2 - Beta(0.5, 0.5)						814.2
U2 - Beta(0.7, 0.3)						817.9
U2 - Beta(1, 1)						813
U3 - Beta(0.3, 0.7)						797.5
U3 - Beta(0.5, 0.5)						813.3
U3 - Beta(0.7, 0.3)						814.9
U3 - Beta(1, 1)						814

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	87.8	9	0.1	0	0	3.1
U2 - Beta(0.5, 0.5)	69.9	25.6	0.7	0	0	3.8
U2 - Beta(0.7, 0.3)	70.2	25	1.7	0	0	3.1
U2 - Beta(1, 1)	69.5	26	1.1	0	0	3.4
U3 - Beta(0.3, 0.7)	88.3	8.7	0	0	0	3
U3 - Beta(0.5, 0.5)	72.9	23.2	0.5	0	0	3.4
U3 - Beta(0.7, 0.3)	70.3	25.2	1.3	0	0	3.2
U3 - Beta(1, 1)	73.9	22.5	0.3	0	0	3.3
Mean administrations						
U2 - Beta(0.3, 0.7)	26.4	2.9	0	0	0	-
U2 - Beta(0.5, 0.5)	20.1	8.2	0.9	0.1	0	-
U2 - Beta(0.7, 0.3)	18.8	8.6	1.7	0.4	0	-
U2 - Beta(1, 1)	20	8.3	0.9	0.1	0	-
U3 - Beta(0.3, 0.7)	26.5	2.9	0	0	0	-
U3 - Beta(0.5, 0.5)	20.7	8	0.6	0.1	0	-
U3 - Beta(0.7, 0.3)	18.9	8.8	1.4	0.2	0	-
U3 - Beta(1, 1)	20.8	8	0.5	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7.5	1.2	0	0	0	-
U2 - Beta(0.5, 0.5)	5	3.1	0.3	0	0	-
U2 - Beta(0.7, 0.3)	4.7	3.1	0.5	0.1	0	-
U2 - Beta(1, 1)	5	3.2	0.3	0	0	-
U3 - Beta(0.3, 0.7)	7.5	1.2	0	0	0	-
U3 - Beta(0.5, 0.5)	5.2	3.1	0.2	0	0	-
U3 - Beta(0.7, 0.3)	4.6	3.3	0.5	0.1	0	-
U3 - Beta(1, 1)	5.3	3.1	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1139.4
U2 - Beta(0.5, 0.5)						1149.2
U2 - Beta(0.7, 0.3)						1162.2
U2 - Beta(1, 1)						1151.1
U3 - Beta(0.3, 0.7)						1140.5
U3 - Beta(0.5, 0.5)						1149.4
U3 - Beta(0.7, 0.3)						1156
U3 - Beta(1, 1)						1149.3

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.2	18.6	25.4	44.8	1.6	0.4
U2 - Beta(0.5, 0.5)	14.3	17.2	18.6	43.1	6.4	0.4
U2 - Beta(0.7, 0.3)	14.9	17.4	19.1	42	6.1	0.5
U2 - Beta(1, 1)	14.8	15.4	18.9	44.4	6.1	0.4
U3 - Beta(0.3, 0.7)	17.3	23.4	23.9	34.6	0.4	0.4
U3 - Beta(0.5, 0.5)	19.5	18.9	16.1	40.6	4.4	0.5
U3 - Beta(0.7, 0.3)	21	19.3	16.5	37.3	5.5	0.4
U3 - Beta(1, 1)	20.7	17.8	16.8	39.7	4.6	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6.1	8.3	8.2	6.9	0.4	-
U2 - Beta(0.5, 0.5)	4.7	6.1	7	9.9	2.3	-
U2 - Beta(0.7, 0.3)	4.9	5.9	7	9	3.2	-
U2 - Beta(1, 1)	4.7	6.2	6.9	9.9	2.3	-
U3 - Beta(0.3, 0.7)	7.1	9.1	7.6	5.9	0.2	-
U3 - Beta(0.5, 0.5)	4.8	6.5	6.9	9.7	2.1	-
U3 - Beta(0.7, 0.3)	5.3	6.3	6.7	9	2.6	-
U3 - Beta(1, 1)	5.1	6.5	6.8	9.6	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	4.2	3.1	2.4	0.2	-
U2 - Beta(0.5, 0.5)	0.7	3.4	2.3	3.4	0.8	-
U2 - Beta(0.7, 0.3)	0.8	3.4	2.1	3.1	1	-
U2 - Beta(1, 1)	0.7	3.5	2.3	3.3	0.8	-
U3 - Beta(0.3, 0.7)	1.9	4.5	2.9	2.1	0.1	-
U3 - Beta(0.5, 0.5)	0.8	3.6	2.3	3.3	0.7	-
U3 - Beta(0.7, 0.3)	1	3.6	2.1	3	0.9	-
U3 - Beta(1, 1)	0.9	3.6	2.2	3.2	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021.7
U2 - Beta(0.5, 0.5)						1064.2
U2 - Beta(0.7, 0.3)						1071.6
U2 - Beta(1, 1)						1065.7
U3 - Beta(0.3, 0.7)						1012.7
U3 - Beta(0.5, 0.5)						1060.5
U3 - Beta(0.7, 0.3)						1063.6
U3 - Beta(1, 1)						1059

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.9	78.8	7.8	0.5	0	0
U2 - Beta(0.5, 0.5)	1.5	54.5	31.8	11.6	0.6	0
U2 - Beta(0.7, 0.3)	3.8	34.1	29.4	26.8	5.9	0
U2 - Beta(1, 1)	1.3	52.7	34	11.4	0.6	0
U3 - Beta(0.3, 0.7)	15.2	79.8	4.7	0.3	0	0
U3 - Beta(0.5, 0.5)	2.3	59.9	27.7	9.6	0.5	0
U3 - Beta(0.7, 0.3)	3.3	42.1	29.5	22.5	2.6	0
U3 - Beta(1, 1)	2	59.6	28.8	9.2	0.4	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	18.6	1.4	0.1	0	-
U2 - Beta(0.5, 0.5)	4.1	16.2	7.3	2.3	0.2	-
U2 - Beta(0.7, 0.3)	3.7	7.9	8.2	7.7	2.5	-
U2 - Beta(1, 1)	4	16.1	7.5	2.3	0.2	-
U3 - Beta(0.3, 0.7)	10.2	18.8	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	4.1	17.3	6.5	1.9	0.1	-
U3 - Beta(0.7, 0.3)	4	10.7	8	5.9	1.4	-
U3 - Beta(1, 1)	4.2	17.1	6.7	1.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	6.7	0.5	0	0	-
U2 - Beta(0.5, 0.5)	0.4	6	2.3	0.7	0.1	-
U2 - Beta(0.7, 0.3)	0.2	3.4	2.3	2.1	0.8	-
U2 - Beta(1, 1)	0.3	6	2.3	0.7	0.1	-
U3 - Beta(0.3, 0.7)	3.1	6.8	0.3	0	0	-
U3 - Beta(0.5, 0.5)	0.4	6.4	2	0.6	0	-
U3 - Beta(0.7, 0.3)	0.3	4.3	2.4	1.8	0.4	-
U3 - Beta(1, 1)	0.4	6.4	2.1	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1091.9
U2 - Beta(0.5, 0.5)						1132.7
U2 - Beta(0.7, 0.3)						1160.9
U2 - Beta(1, 1)						1132.3
U3 - Beta(0.3, 0.7)						1090.3
U3 - Beta(0.5, 0.5)						1130.7
U3 - Beta(0.7, 0.3)						1149.6
U3 - Beta(1, 1)						1130.1
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.3	11.1	52.3	30.3	2.9	0.1
U2 - Beta(0.5, 0.5)	9.2	11.5	35.7	35.3	8.1	0.2
U2 - Beta(0.7, 0.3)	10.2	16.1	34.7	31.4	7.4	0.2
U2 - Beta(1, 1)	9.9	13.3	34.6	34.4	7.6	0.2
U3 - Beta(0.3, 0.7)	5.4	16.3	51.7	25.2	1.3	0.1
U3 - Beta(0.5, 0.5)	13	17.3	36.4	28.5	4.7	0.1
U3 - Beta(0.7, 0.3)	14.1	20	34.4	26.4	4.9	0.2
U3 - Beta(1, 1)	14.8	18.9	34.3	28.2	3.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	7.4	12.2	5.3	0.7	-
U2 - Beta(0.5, 0.5)	3.5	4.7	9.1	9	3.6	-
U2 - Beta(0.7, 0.3)	3.6	4.6	8.7	8.9	4.1	-
U2 - Beta(1, 1)	3.5	4.8	9.1	9	3.6	-
U3 - Beta(0.3, 0.7)	4.6	8	12.1	4.8	0.4	-
U3 - Beta(0.5, 0.5)	3.6	4.9	9.5	8.9	3.1	-
U3 - Beta(0.7, 0.3)	3.8	5.1	9	8.7	3.5	-
U3 - Beta(1, 1)	3.8	5.3	9.4	8.7	2.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.9	4.7	2	0.3	-
U2 - Beta(0.5, 0.5)	0.2	3	3	3.4	1.3	-
U2 - Beta(0.7, 0.3)	0.3	3	2.8	3.4	1.3	-
U2 - Beta(1, 1)	0.2	3	3	3.4	1.2	-
U3 - Beta(0.3, 0.7)	0.8	4.2	4.6	1.8	0.2	-
U3 - Beta(0.5, 0.5)	0.3	3.1	3.2	3.3	1.1	-
U3 - Beta(0.7, 0.3)	0.3	3.2	2.9	3.2	1.2	-
U3 - Beta(1, 1)	0.3	3.3	3.1	3.3	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016.7
U2 - Beta(0.5, 0.5)						1049.7
U2 - Beta(0.7, 0.3)						1055.6
U2 - Beta(1, 1)						1049.9
U3 - Beta(0.3, 0.7)						1010.7
U3 - Beta(0.5, 0.5)						1048
U3 - Beta(0.7, 0.3)						1050.4
U3 - Beta(1, 1)						1046.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	6.2	22.2	60.2	6.7	0.1
U2 - Beta(0.5, 0.5)	6.6	7.9	16.3	53.2	15.9	0.1
U2 - Beta(0.7, 0.3)	8.1	9.9	15.2	50.4	16.3	0.1
U2 - Beta(1, 1)	7.4	8.8	15.3	52.4	16	0.1
U3 - Beta(0.3, 0.7)	8.3	4.6	24.7	58.3	4	0.1
U3 - Beta(0.5, 0.5)	12.6	7.9	13.9	52.7	12.8	0.1
U3 - Beta(0.7, 0.3)	12.9	10.5	13.6	47.9	15	0.1
U3 - Beta(1, 1)	14.8	9.5	12.8	50.9	11.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.9	9.2	11	1.1	-
U2 - Beta(0.5, 0.5)	3.3	3.9	6.6	11.7	4.5	-
U2 - Beta(0.7, 0.3)	3.4	3.8	6.4	10.5	5.9	-
U2 - Beta(1, 1)	3.4	3.9	6.6	11.6	4.5	-
U3 - Beta(0.3, 0.7)	3.8	5	9.5	10.8	0.8	-
U3 - Beta(0.5, 0.5)	3.4	4	6.7	11.8	4.1	-
U3 - Beta(0.7, 0.3)	3.6	4.1	6.5	10.6	5.2	-
U3 - Beta(1, 1)	3.6	4.2	6.6	11.7	3.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3	3.5	3.8	0.4	-
U2 - Beta(0.5, 0.5)	0.1	2.8	2	4.1	1.3	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.9	3.8	1.6	-
U2 - Beta(1, 1)	0.2	2.8	1.9	4.1	1.3	-
U3 - Beta(0.3, 0.7)	0.4	3.1	3.6	3.8	0.3	-
U3 - Beta(0.5, 0.5)	0.2	2.8	2	4.2	1.2	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.9	3.8	1.5	-
U3 - Beta(1, 1)	0.3	3	1.9	4.1	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039
U2 - Beta(0.5, 0.5)						1081.5
U2 - Beta(0.7, 0.3)						1087
U2 - Beta(1, 1)						1080.2
U3 - Beta(0.3, 0.7)						1033.8
U3 - Beta(0.5, 0.5)						1078.7
U3 - Beta(0.7, 0.3)						1080.5
U3 - Beta(1, 1)						1075.2
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	18.3	56.4	10.2	0.4	4.6
U2 - Beta(0.5, 0.5)	13.2	18.8	49.4	12.5	1.5	4.6
U2 - Beta(0.7, 0.3)	14.2	21.1	45.9	12.4	1.6	4.8
U2 - Beta(1, 1)	14.6	19.6	46.3	13.2	1.6	4.7
U3 - Beta(0.3, 0.7)	12.6	20	51.7	9.9	0.3	5.5
U3 - Beta(0.5, 0.5)	17.4	20.9	43.6	12.4	0.9	4.8
U3 - Beta(0.7, 0.3)	18.9	20.1	43.2	12.4	0.9	4.5
U3 - Beta(1, 1)	18.9	20	41.5	14.2	0.8	4.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	8.2	12.3	3.2	0.2	-
U2 - Beta(0.5, 0.5)	5.6	7.7	10.2	5	1.1	-
U2 - Beta(0.7, 0.3)	5.6	7.6	9.7	5.2	1.4	-
U2 - Beta(1, 1)	5.7	7.7	10.2	5	1	-
U3 - Beta(0.3, 0.7)	6.2	8.6	11.8	2.8	0.1	-
U3 - Beta(0.5, 0.5)	6	7.8	10.1	4.8	0.9	-
U3 - Beta(0.7, 0.3)	6.1	7.8	9.5	5.1	1.1	-
U3 - Beta(1, 1)	6.3	7.8	9.8	4.8	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	4.1	4.3	1.2	0.1	-
U2 - Beta(0.5, 0.5)	1.1	4.1	3.3	1.9	0.4	-
U2 - Beta(0.7, 0.3)	1.1	4.1	3.1	1.9	0.5	-
U2 - Beta(1, 1)	1.2	4.1	3.2	1.9	0.4	-
U3 - Beta(0.3, 0.7)	1.5	4.2	4.1	1.1	0	-
U3 - Beta(0.5, 0.5)	1.3	4.2	3.2	1.8	0.3	-
U3 - Beta(0.7, 0.3)	1.3	4.2	3	1.8	0.4	-
U3 - Beta(1, 1)	1.5	4.2	3.1	1.8	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1023.6
U2 - Beta(0.5, 0.5)						1037
U2 - Beta(0.7, 0.3)						1040.2
U2 - Beta(1, 1)						1035.8
U3 - Beta(0.3, 0.7)						1017.6
U3 - Beta(0.5, 0.5)						1033
U3 - Beta(0.7, 0.3)						1034.2
U3 - Beta(1, 1)						1031

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.3	77.3	8.9	0.4	0	0.1
U2 - Beta(0.5, 0.5)	4.9	55.1	35.3	4.4	0.2	0.1
U2 - Beta(0.7, 0.3)	6.9	49.8	31.9	10.9	0.4	0.1
U2 - Beta(1, 1)	5.9	55.4	34	4.6	0	0.1
U3 - Beta(0.3, 0.7)	15.8	77.2	6.7	0.2	0	0.1
U3 - Beta(0.5, 0.5)	6.4	61.7	28.3	3.5	0	0.1
U3 - Beta(0.7, 0.3)	7.6	51.4	32.1	8.6	0.2	0.1
U3 - Beta(1, 1)	7.4	62.8	26.6	3.1	0	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	18.3	1.7	0.1	0	-
U2 - Beta(0.5, 0.5)	4.4	15.5	8.2	1.7	0.2	-
U2 - Beta(0.7, 0.3)	4.4	10.9	9.1	4.5	1	-
U2 - Beta(1, 1)	4.5	15.5	8.2	1.7	0.1	-
U3 - Beta(0.3, 0.7)	10.2	18.4	1.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4.8	16.5	7.2	1.4	0.1	-
U3 - Beta(0.7, 0.3)	4.6	12.8	8.7	3.4	0.5	-
U3 - Beta(1, 1)	4.8	16.6	7.1	1.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	6.8	0.5	0	0	-
U2 - Beta(0.5, 0.5)	0.5	5.8	2.4	0.6	0.1	-
U2 - Beta(0.7, 0.3)	0.5	4.3	2.5	1.4	0.3	-
U2 - Beta(1, 1)	0.5	5.8	2.4	0.6	0.1	-
U3 - Beta(0.3, 0.7)	3.2	6.8	0.4	0	0	-
U3 - Beta(0.5, 0.5)	0.6	6.3	2.1	0.5	0	-
U3 - Beta(0.7, 0.3)	0.6	4.9	2.5	1.1	0.2	-
U3 - Beta(1, 1)	0.7	6.3	2	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1080.9
U2 - Beta(0.5, 0.5)						1135.7
U2 - Beta(0.7, 0.3)						1155.3
U2 - Beta(1, 1)						1134.9
U3 - Beta(0.3, 0.7)						1078.6
U3 - Beta(0.5, 0.5)						1129.6
U3 - Beta(0.7, 0.3)						1146.8
U3 - Beta(1, 1)						1128.6
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.4	35.7	10.4	1.6	0	35.9
U2 - Beta(0.5, 0.5)	16.9	36.7	9.5	2.1	0.3	34.5
U2 - Beta(0.7, 0.3)	17.2	36.7	10.6	1.6	0.2	33.7
U2 - Beta(1, 1)	16.9	37.3	9.3	2.4	0.2	33.9
U3 - Beta(0.3, 0.7)	17.6	36.7	8.7	1.3	0	35.7
U3 - Beta(0.5, 0.5)	20.1	35.9	8	1.7	0.1	34.2
U3 - Beta(0.7, 0.3)	17.9	37.6	8.9	1.3	0.2	34.1
U3 - Beta(1, 1)	20.2	36.5	6.8	2	0.1	34.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	11.2	3.8	1	0	-
U2 - Beta(0.5, 0.5)	9.8	9.9	4.2	1.9	0.3	-
U2 - Beta(0.7, 0.3)	9.6	9.8	4.3	2.1	0.4	-
U2 - Beta(1, 1)	9.6	10.1	4.3	1.9	0.3	-
U3 - Beta(0.3, 0.7)	10.1	11.3	3.5	0.9	0	-
U3 - Beta(0.5, 0.5)	10	10	4.1	1.8	0.3	-
U3 - Beta(0.7, 0.3)	9.8	10	4.1	2	0.3	-
U3 - Beta(1, 1)	10.1	10.1	3.9	1.8	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	5.3	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	2.5	4.9	1.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.4	4.9	1.4	0.8	0.1	-
U2 - Beta(1, 1)	2.4	5	1.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.7	5.3	1.3	0.4	0	-
U3 - Beta(0.5, 0.5)	2.5	5	1.3	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.5	5	1.3	0.8	0.1	-
U3 - Beta(1, 1)	2.6	5	1.2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						885.5
U2 - Beta(0.5, 0.5)						904.5
U2 - Beta(0.7, 0.3)						909.5
U2 - Beta(1, 1)						905.3
U3 - Beta(0.3, 0.7)						880.6
U3 - Beta(0.5, 0.5)						900.3
U3 - Beta(0.7, 0.3)						906.7
U3 - Beta(1, 1)						903.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.4	44.8	10	0.2	0.1	27.5
U2 - Beta(0.5, 0.5)	18.1	47	7.8	0.2	0	26.9
U2 - Beta(0.7, 0.3)	17.1	47.1	8.9	0.3	0	26.6
U2 - Beta(1, 1)	17.2	46.1	8.9	0.3	0	27.5
U3 - Beta(0.3, 0.7)	19.8	45.9	7.6	0.2	0	26.5
U3 - Beta(0.5, 0.5)	20.2	45.9	7.1	0.4	0	26.4
U3 - Beta(0.7, 0.3)	19.9	45	7.6	0.5	0	27
U3 - Beta(1, 1)	20.8	43.9	7.5	0.6	0	27.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	12.9	4	0.8	0.1	-
U2 - Beta(0.5, 0.5)	9.4	11.6	4.4	1.7	0.3	-
U2 - Beta(0.7, 0.3)	9.3	11.4	4.4	2	0.3	-
U2 - Beta(1, 1)	9.2	11.7	4.4	1.8	0.3	-
U3 - Beta(0.3, 0.7)	9.8	13.2	3.5	0.7	0	-
U3 - Beta(0.5, 0.5)	9.7	11.6	4.2	1.6	0.2	-
U3 - Beta(0.7, 0.3)	9.6	11.4	4.2	1.8	0.3	-
U3 - Beta(1, 1)	9.7	11.7	4.2	1.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	6	1.5	0.4	0	-
U2 - Beta(0.5, 0.5)	2.3	5.7	1.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.3	5.7	1.3	0.8	0.1	-
U2 - Beta(1, 1)	2.2	5.7	1.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.7	6	1.3	0.3	0	-
U3 - Beta(0.5, 0.5)	2.4	5.7	1.3	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.4	5.7	1.3	0.8	0.1	-
U3 - Beta(1, 1)	2.4	5.7	1.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						925.4
U2 - Beta(0.5, 0.5)						940
U2 - Beta(0.7, 0.3)						942.7
U2 - Beta(1, 1)						941
U3 - Beta(0.3, 0.7)						923.4
U3 - Beta(0.5, 0.5)						938.7
U3 - Beta(0.7, 0.3)						936.1
U3 - Beta(1, 1)						938.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.